

CONVERGENCE RATES FOR THE RMSPROP OPTIMIZER WITH FULL CONTROL OF THE HYPERPARAMETERS

STEFFEN DEREICH AND ARNULF JENTZEN

ABSTRACT. *Stochastic gradient descent* (SGD) optimization methods are the standard methods to train *artificial intelligence* (AI) systems. Often the standard SGD method is not the used training algorithm but instead one considers suitable adaptive variants of SGD in which the learning rates are for each network parameter separately adjusted adaptively during the training process. Popular adaptive SGD methods include the RMSprop, the Adam, and the AdamW optimizers, where the adaptivity parts in Adam and AdamW basically just coincide with RMSprop.

Such adaptive methods involve several *hyperparameters* including the regularization parameter ε (which ensures that one does not divide by zero during the training process and is often chosen to be very close to zero such as 10^{-8} in PyTorch by default) as well as the second moment decay parameter β (which is often chosen to be very close to 1 such as 0.99 (RMSprop) and 0.999 (Adam and AdamW) in PyTorch by default). Despite the extensive literature on the theoretical analysis of such adaptive methods, it remains an open research problem to provide error estimates for such methods with the error constants being provably *not exploding but uniformly bounded with the respect to the hyperparameters* approaching the critical regions, even in the situation of convex stochastic optimization problems (SOPs).

It is the key contribution of this work to essentially solve this problem for the RMSprop optimizer. Specifically, our main result provides bounds the expectation of the stopped evaluation of the objective function at the RMSprop optimization process from above by the sum of an initialization term that decays exponentially in the training time, a stochastic approximation remainder of order γ_n , and a memory error of order $(1 - \beta)^2$ with the error constants being uniformly controlled over all admissible choices of the step sizes $(\gamma_n)_{n \in \mathbb{N}}$, the second moment decay parameter β and the regularization parameter $\varepsilon \in [0, 1]$. In particular, we emphasize that our error analysis also directly covers the *unregularized regime* in which the regularity parameter vanishes $\varepsilon = 0$. The non-asymptotic optimization error estimates that we establish hold not just for all sufficiently large number of gradient steps but instead hold for every gradient step $n \in \mathbb{N} = \{1, 2, 3, \dots\}$ (starting at the very first gradient step) with *all error constants being explicitly specified*.

The key innovative new feature in the proof of our error analysis are suitable *inverse moment bounds* for the second moment process in the RMSprop method. We also illustrate the general theory of this work in the case of several example SOPs.

2020 *Mathematics Subject Classification.* Primary 65K10; Secondary 90C15, 90C26, 60G42.

Key words and phrases. RMSprop, adaptive, stochastic gradient descent, SGD, stochastic approximation, stochastic optimization, Kurdyka–Łojasiewicz inequality, hyperparameters.

CONTENTS

1. Introduction	2
1.1. The RMSprop method	3
1.2. Main result: Error analysis for RMSprop with full control of the hyperparameters	4
1.3. Example: RMSprop for least squares with full control of the hyperparameters	5
1.4. Short literature review and key novelty of this work	6
1.5. Overview and structure of this work	7
1.6. Use of large language models	7
2. The abstract recursion estimate in Proposition 2.1	7
3. Analysis of the property in item (iii) of Proposition 2.1	10
4. Inverse moment estimates for the second moment process $(v_n)_{n \in \mathbb{N}_0}$ in the RMSprop method	12
5. Technical estimates for the property in item (ii) of Proposition 2.1	17
6. Initialisation estimates	19
7. Combining the estimates before and after the coordinate startup times	22
8. Criteria and examples for regular minibatch innovations	27
8.1. A small-ball criterion for regularity	27
8.2. Strong convexity in the data variable	27
8.3. Nonlinear regression	30
References	34

1. INTRODUCTION

Artificial Intelligence (AI) systems are trained by means of *stochastic gradient descent* (SGD) optimization methods [33]. The standard SGD optimization method is often not the employed training scheme for large scale AI systems but instead one often considers suitable adaptive variants of standard SGD such as the *root mean square propagation* SGD (RMSprop) [35], the *adaptive moment estimation* SGD (Adam) [25], and the *Adam with decoupled weight decay* (AdamW) [28] optimizers [21].

In the RMSprop method essentially every coordinate of the update is divided by the square root of a geometric average of the square of all previously seen stochastic gradients and the adaptivity parts in Adam and AdamW are essentially nothing else but the RMSprop method. Such adaptive SGD optimization methods are based on several *hyperparameters* in the optimizers including

- the sequence of learning rates $(\gamma_n)_{n \in \mathbb{N}}$ (the sequence of step sizes),
- the regularization parameter ε (which ensures that one does not divide by zero in each update) as well as
- the second moment decay parameter β (which, loosely speaking, models how much weight the recent stochastic gradients get within the geometric average of the square of all previously seen stochastic gradients).

The regularization parameter ε is often chosen to be very close to zero such as 10^{-8} in PyTorch by default and the second moment decay parameter β is often chosen to be very close to 1 such as 0.99 (RMSprop) and 0.999 (Adam and AdamW) in PyTorch by default.

Despite the already existing extensive literature on the theoretical analysis of such adaptive SGD optimization methods and the high importance of such methods, it remains an open problem of research to prove (or disprove) rigorous error estimates for such optimizers with the error constants being provably not exploding but *uniformly bounded with the respect to the hyperparameters* approaching the critical regions, even in the situation of strongly convex *stochastic optimization problems* (SOPs).

It is the key contribution of this work to essentially solve this problem for the RMSprop optimizer. Specifically, in our main result in Theorem 1.1 below we bound the expectation of a stopped evaluation of the objective

function at the [RMSprop](#) optimization process (which controls the squared mean square strong optimization error in the special case of strongly convex [SOPs](#)) from above by the sum of

- an initialization term that decays exponentially in the training time $t_n = \sum_{k=1}^n \gamma_k$,
- a stochastic approximation remainder of order γ_n , and
- a memory error of order $(1 - \beta)^2$

with the error constants being *uniformly controlled* over all admissible choices of the step sizes $(\gamma_n)_{n \in \mathbb{N}}$, the second moment decay parameter β and the regularization parameter $\varepsilon \in [0, 1]$. Particularly, we highlight that our error analysis also directly covers the *unregularized regime* in which the regularity parameter completely vanishes $\varepsilon = 0$.

1.1. The RMSprop method. In order to formulate [Theorem 1.1](#), we present the considered mathematical setup and we briefly recall the [RMSprop](#) method.

Let $d \in \mathbb{N} = \{1, 2, 3, \dots\}$, let $(\mathcal{U}, \mathbb{U})$ be a measurable space, let $X: \mathbb{R}^d \times \mathcal{U} \rightarrow \mathbb{R}^d$ be product measurable, let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and let U be an $\mathcal{U}$ -valued random variable. We call the pair (X, U) an *innovation*. The [RMSprop](#) algorithm driven by (X, U) depends on the following additional parameters:

- $\beta \in [0, 1)$ (*second moment decay parameter*), $\varepsilon \in [0, \infty)$ (*regularization parameter*),
- a non-increasing $(0, \infty)$ -valued sequence $(\gamma_n)_{n \in \mathbb{N}}$ (*step-sizes/learning rates*), and
- $\theta_0 \in \mathbb{R}^d$ (*initial state/value*).

Let U_n , $n \in \mathbb{N}$, be independent copies of U and let $(\mathcal{F}_n)_{n \in \mathbb{N}_0}$ be the filtration generated by these copies, that is, for every $n \in \mathbb{N}_0$ let

$$\mathcal{F}_n = \sigma(U_1, U_2, \dots, U_n). \quad (1)$$

The associated [RMSprop](#) algorithm is the $(\mathcal{F}_n)_{n \in \mathbb{N}_0}$ -adapted process $(\theta_n)_{n \in \mathbb{N}_0}$ defined recursively, for every $n \in \mathbb{N}$, by

$$\theta_n = \theta_{n-1} + \gamma_n [\varepsilon + \sqrt{v_n}]^{-1} X(\theta_{n-1}, U_n), \quad (2)$$

where all operations are applied coordinatewise to vectors and where, for every $n \in \mathbb{N}$, we have that

$$v_n = \frac{1 - \beta}{1 - \beta^n} \sum_{k=0}^{n-1} \beta^k X(\theta_{n-1-k}, U_{n-k})^2. \quad (3)$$

Equivalently, we have for every $n \in \mathbb{N}$ that

$$v_n = \frac{\beta(1 - \beta^{n-1})}{1 - \beta^n} v_{n-1} + \frac{1 - \beta}{1 - \beta^n} X(\theta_{n-1}, U_n)^2. \quad (4)$$

where θ_0 is the predetermined initial state and where $v_0 = 0 \in \mathbb{R}^d$. If $\varepsilon = 0$, all coordinatewise normalized innovations are understood with the convention

$$0^{-1}0 = \frac{0}{0} = 0. \quad (5)$$

We call the mapping $f_X: \mathbb{R}^d \rightarrow \mathbb{R}^d$ given, for every $\theta \in \mathbb{R}^d$, by $f_X(\theta) = \mathbb{E}[X(\theta, U)]$ the *vector field associated with the innovation* (X, U) .

A key hypothesis in our error analysis is the following *non-degeneracy assumption*, which we formalize in [Definition 4.1](#) below. For suitable $c, q > 0$ we have for every $i \in \{1, 2, \dots, d\}$ that the Laplace transform of $X^{(i)}(\theta, U)^2$ satisfies

$$\mathbb{E}[e^{-\lambda X^{(i)}(\theta, U)^2}] \leq \exp(-\min\{c\lambda, q\}). \quad (6)$$

Loosely speaking, this hypothesis ensures for every component of the innovation that the probability that this component vanishes is not too large.

In our error analysis result for the [RMSprop](#) method in [Theorem 1.1](#) below we employ this assumption together with locally L^∞ -bounded innovations and a local *Polyak–Łojasiewicz* (PL) inequality (see [\(11\)](#) below). The additional condition in [\(11\)](#) prevents abrupt decreases of the step-size when measured with respect to the

training time $t_n = \sum_{k=1}^n \gamma_k$ for every $n \in \mathbb{N}_0$. The condition in (11) is satisfied, in particular, in the situation of the usual polynomial learning rate schedules.

1.2. Main result: Error analysis for RMSprop with full control of the hyperparameters. The following non-asymptotic error analysis result for the **RMSprop** method is a consequence of the general recursive error estimate in [Theorem 7.1](#), which is stated and proved in [Section 7](#) below. In [Theorem 1.2](#) in [Subsection 1.3](#) below we illustrate the conclusion of [Theorem 1.1](#) in the example of linear least squares (example application of [Theorem 1.1](#)).

Theorem 1.1 (**Error analysis for RMSprop with full control of the hyperparameters**). *Let $\rho, c, q, C_X, L_f, C_{\text{Loj}}, C_{\text{dec}} \in (0, \infty)$. Set*

$$\eta = [2C_{\text{Loj}}(C_X + 1)]^{-1}. \quad (7)$$

Then there exist $C_0, C_1 \in (0, \infty)$ such that the following is true. Let $\beta \in [1/2, 1)$, $\varepsilon \in [0, 1]$ satisfy

$$q + 3 \ln \beta \geq \rho. \quad (8)$$

Let $V \subseteq \mathbb{R}^d$ be measurable, let (X, U) be (c, q) -regular on V in the sense of [Definition 4.1](#) and assume for every $\theta \in V$, $i \in \{1, \dots, d\}$ that

$$\|X^{(i)}(\theta, U)\|_{L^\infty} \leq C_X. \quad (9)$$

Let $F: \mathbb{R}^d \rightarrow [0, \infty)$ have a globally L_f -Lipschitz continuous derivative and assume for every $\theta \in V$ that

$$f_X(\theta) = -\nabla F(\theta) \quad \text{and} \quad F(\theta) \leq C_{\text{Loj}} |f_X(\theta)|^2. \quad (10)$$

Let $(\gamma_n)_{n \in \mathbb{N}}$ be decreasing and $(0, C_{\text{Loj}}(C_X + \varepsilon)]$ -valued and assume for every $j, n \in \mathbb{N}$ with $1 < j \leq n$ that

$$\gamma_j \leq \gamma_n C_{\text{dec}} \exp\left(\frac{\eta}{2} \sum_{k=j+1}^n \gamma_k\right). \quad (11)$$

*Let $\theta_0 \in \mathbb{R}^d$ and let $(\theta_n)_{n \in \mathbb{N}_0}$ be the **RMSprop** algorithm driven by (X, U) with parameter tuple $(\beta, \varepsilon, (\gamma_n)_{n \in \mathbb{N}}, \theta_0)$ defined above. Define $\tau = \inf\{n \in \mathbb{N}_0 : \theta_n \notin V\}$. Then one has for every $n \in \mathbb{N}$ that*

$$\mathbb{E}[F(\theta_n) \mathbf{1}_{\{\tau \geq n\}}] \leq \exp(-\eta \sum_{k=2}^n \gamma_k) [F(\theta_0)^{1/2} + \gamma_1 (2^{-1} L_f d)^{1/2}]^2 + C_0 \gamma_n + C_1 (1 - \beta)^2. \quad (12)$$

[Theorem 1.1](#) follows from an application of the general recursive error estimate in [Theorem 7.1](#), which is stated and proved in [Section 7](#) below. We refer to the end of [Section 7](#) for the detailed proof of [Theorem 1.1](#).

In (12) we bound the expectation of the τ -stopped evaluation of the objective function at the **RMSprop** optimization process from above by the sum

- (i) of the initialization term that converges to zero as the training time $t_n = \sum_{k=1}^n \gamma_k$ converges to infinity in the number of **RMSprop** steps n ,
- (ii) of the term $C_0 \gamma_n$ (the usual stochastic approximation error) that converges to zero as the number of **RMSprop** steps n goes to infinity, and
- (iii) of the term $C_1 (1 - \beta)^2$ that converges to zero as the second moment decay parameter β approaches 1.

The convergence speed $C_0 \gamma_n$ according to [item \(ii\)](#) is the usual stochastic approximation error term that also appears in the mean square error analysis of the standard **SGD** method and is optimal and cannot be improved (cf., for instance, [\[23\]](#)). In the case of **Adam** numerical simulations and analytical considerations suggest that the power two in the error term $C_1 (1 - \beta)^2$ can also not be improved (cf., for example, [\[7\]](#))

We highlight that, except of C_0 and C_1 , all constants in [Theorem 1.1](#) are explicitly specified and, actually, even C_0 and C_1 can be explicitly specified (see [Remark 7.2](#) in [Section 7](#) below). In particular, [Remark 7.2](#) shows that C_0 and C_1 in [Theorem 1.1](#) can be chosen as

$$C_0 = 2^{13} C_{\text{dec}} d (L_f + C_X + 1)^7 (\eta^{-1} + 1) \left[\sum_{r=0}^{\infty} \frac{(c^{-3} + 1)(r + 1)^3}{\exp(\min\{\rho, 2^{-4} q\} r)} \right] \quad (13)$$

and

$$C_1 = 2^{13} d L_f C_X^6 \eta^{-2} \left[\sum_{r=0}^{\infty} \frac{c^{-3}(r+1)^3}{\exp(\rho r)} \right]. \quad (14)$$

We refer to (206), (208), and (210) in Remark 7.2 for details. We also emphasize that all of the error constants in (12) are completely independent of $\beta \in [1/2, 1)$ and also completely independent of $\varepsilon \in [0, 1]$ and we highlight that even in the *unregularized regime* $\varepsilon = 0$ the conclusion of (12) remains valid under the zero-over-zero convention in (5) above.

In Theorem 1.1 the initial value of the RMSprop optimization process is a purely deterministic vector $\theta_0 \in \mathbb{R}^d$ instead of an $\mathbb{R}^d$ -valued random variable. However, under the assumption that this $\mathbb{R}^d$ -valued random variable is independent from the driving random variables U_n , $n \in \mathbb{N}$, we have that Theorem 1.1 directly implies the corresponding statement but with a stochastic initial state of the RMSprop process as the error constants and parameters η , C_0 , C_1 , L_f , and d in (12) are all completely independent of the initial value θ_0 ,

We also point out that in Theorem 1.1 we do not require that the learning rates γ_n , $n \in \mathbb{N}$, converge to zero and the inequality in (12) holds true even if the initial learning rates $(\gamma_n)_{n \in \mathbb{N}}$ are bounded from above by $C_{\text{Loj}} C_X$ and non-increasing, even though it is well-known that RMSprop and other SGD optimization methods fail to converge if the learning rates do not converge to zero [8].

We also note that the error estimate in (12) in Theorem 1.1 is restricted to the event $\{\tau \geq n\}$ and is thus only applicable until the optimization process $(\theta_n)_{n \in \mathbb{N}_0}$ has left the measurable set V . We think of the set V as a sufficiently large compact ball (see, for instance, (17) in Theorem 1.2 below) and we refer, for example, to [6] and [9, Theorem 2.10] for results in the literature that establish a priori bounds for RMSprop and other related adaptive SGD optimization methods (particularly Adam) which ensure that the considered optimization process remains in a sufficiently large compact ball.

1.3. Example: RMSprop for least squares with full control of the hyperparameters. In the following result, Theorem 1.2 below, we illustrate the conclusion of Theorem 1.1 above in the example of linear least squares. In Section 8 below we show in detail how Theorem 1.1 can be applied to prove Theorem 1.2.

Theorem 1.2 (RMSprop for linear least squares). *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, let $\kappa \in (0, \infty)$, $m, d, M \in \mathbb{N}$, $\vartheta \in \mathbb{R}^d$, let $A \in \mathbb{R}^{m \times d}$ have no zero columns, let $K \subseteq \mathbb{R}^m$ be compact, let $(Y_{n,r})_{(n,r) \in \mathbb{N}^2}$ be K -valued i.i.d. random variables with a bounded Lebesgue density, let $L: \mathbb{R}^d \times \mathbb{R}^m \rightarrow \mathbb{R}$ satisfy for all $\theta \in \mathbb{R}^d$, $y \in \mathbb{R}^m$ that*

$$L(\theta, y) = |A\theta - y|^2 \quad (15)$$

and let $F: \mathbb{R}^d \rightarrow \mathbb{R}$ satisfy for all $\theta \in \mathbb{R}^d$ that $F(\theta) = \mathbb{E}[L(\theta, Y_{1,1})] - \inf_{\vartheta \in \mathbb{R}^d} \mathbb{E}[L(\vartheta, Y_{1,1})]$. Then there exists $\eta \in (0, \infty)$ such that for every $\mathfrak{C} \in (0, \infty)$ there exists $c \in \mathbb{R}$ such that for all $\beta \in [1/2, 1)$, $\varepsilon \in [0, 1]$, all $\mathbb{R}^d$ -valued stochastic processes $(w_n)_{n \in \mathbb{N}_0}$ and $(\theta_n)_{n \in \mathbb{N}_0}$, and every $(0, \mathfrak{C}]$ -valued decreasing $(\gamma_n)_{n \in \mathbb{N}}$ with the property that for every $n \in \mathbb{N}$, $j \in \mathbb{N} \cap [1, n]$, $i \in \{1, 2, \dots, d\}$ it holds that

$$\begin{aligned} w_0 &= 0, & w_n^{(i)} &= \beta w_{n-1}^{(i)} + (1 - \beta) \left[M^{-1} \sum_{r=1}^M (\nabla_{\theta_i} L)(\theta_{n-1}, Y_{n,r}) \right]^2, \\ \theta_0 &= \vartheta, & \theta_n^{(i)} &= \theta_{n-1}^{(i)} - \gamma_n [\varepsilon + (1 - \beta^n)^{-1/2} (w_n^{(i)})^{1/2}]^{-1} M^{-1} \sum_{r=1}^M (\nabla_{\theta_i} L)(\theta_{n-1}, Y_{n,r}) \end{aligned} \quad (16)$$

and $\gamma_j \leq \mathfrak{C} \exp(\frac{\eta}{2} \sum_{k=j+1}^n \gamma_k) \gamma_n$ we have for every $n \in \mathbb{N}$ that

$$\mathbb{E} \left[F(\theta_n) \mathbb{1}_{\cap_{j=0}^{n-1} \{|\theta_j| \leq \kappa\}} \right] \leq \exp(-\eta \sum_{k=2}^n \gamma_k) \left(\gamma_1 (d \|A^\top A\|)^{1/2} + |F(\theta_0)|^{1/2} \right)^2 + c \gamma_n + c(1 - \beta)^2. \quad (17)$$

In Section 8 we rigorously show how Theorem 1.2 can be deduced from Theorem 1.1 in Subsection 1.2 above. In addition, in Section 8 we also illustrate the conclusion of Theorem 1.1 in the case of two further classes of example SOPs, namely, risk-sensitive exponential tilting with coordinate gradients that are strongly convex in the data variable and compactly supported nonlinear regression with vector-valued responses.

We point out that in [Theorem 1.2](#) we do not assume that $A^\top A$ is invertible and we note that the objective function $F: \mathbb{R}^d \rightarrow \mathbb{R}$ in [Theorem 1.2](#) is convex but not necessarily *strongly convex*.

1.4. Short literature review and key novelty of this work. Nowadays there are a large number of research works that study adaptive gradient based optimization methods theoretically. In particular,

- we refer, for instance, to [\[4, Theorem 3.1\]](#), [\[16, Subsection 2.4\]](#), [\[24, Subsection 6.4\]](#), [\[27, Theorem 3.6 and Corollary 3.7\]](#), [\[34, Subsection 4.2\]](#), [\[37, Theorem 4.1\]](#), [\[39, Theorem 1, Corollary 2, and Corollary 3\]](#), [\[40, Theorem 1 and Theorem 3\]](#), [\[42, Sections 4 and 5\]](#), and [\[43, Section 3\]](#) for error, complexity, and convergence analyses for the [RMSprop](#) method and minor modifications of it when applied to [SOPs](#),
- we refer, for example, to [\[2, 11, 12, 20, 29\]](#) for works that study [RMSprop](#) in the deterministic/full-batch setting, and
- we refer, for instance, to [\[13, 14, 22\]](#) for lower bounds and non-convergence results to global minimizers for [RMSprop](#) in the training of *artificial neural networks* (ANNs).

For online learning problems we also refer, for example, to [\[31, Theorem 4.1, Section 4, and Assumptions A1–A2\]](#) for upper bounds for the (averaged) regret of a modified variant of the [RMSprop](#) method in which the second moment parameter β is not constant but converging with appropriate speed of convergence to 1 during the training procedure and in which the regularization parameter ε is not constant but converging with appropriate speed of convergence to 0 during the training procedure. We also refer, for instance, to [\[30, Theorem 4.2, Theorem C.2, and Theorem C.5\]](#) for approximation results for [RMSprop](#) applied to [SOPs](#) using solution processes of *stochastic differential equations* (SDEs).

Furthermore, we refer, for example, to [\[1, 3, 5, 7, 10, 19, 26, 32, 36, 38, 41\]](#) for minimized/ergodic/non-ergodic error, gradient, and complexity estimates for the [Adam](#) optimizer (which is basically nothing else but the combination of [RMSprop](#) and momentum) and modified variants of it and we refer, for instance, to [\[7, 18, 32\]](#) for non-convergence results for [Adam](#) due to an *internal error* of the optimizer. [RMSprop](#) also admits such an *internal error* and this is the reason why the additional error term $C_1(1 - \beta)^2$ in (12) in [Theorem 1.1](#) above can, in general, not be avoided. We also refer, for example, to [\[17\]](#) and the references therein for error estimates for other related adaptive [SGD](#) methods such as the *adaptive gradient SGD* ([Adagrad](#)) optimizer [\[15\]](#). Further works on the theoretical analysis of [SGD](#) optimization methods can, for instance, also be found in [\[7, Section 1.3\]](#).

The **key innovative new contribution** of this work is that we establish convergence rates and error estimates for [RMSprop with full control of the hyperparameters](#). Specifically, to the best of our knowledge, [Theorem 1.1](#) is the first result in the scientific literature that establishes convergence of the [RMSprop](#) method for [SOPs with the error constants being uniformly controlled with respect to the hyperparameters](#), particularly, with respect to the second moment decay parameter β and the regularization parameter ε . This work even covers the completely *unregularized case* $\varepsilon = 0$.

This kind of error analysis is very relevant for the application of adaptive optimizers in practically relevant regimes as if the error constants in error estimates are not uniformly bounded with respect to the hyperparameters but instead grow, for example, quadratically in the reciprocal ε^{-1} of the regularization parameter ε and if the regularization parameter is chosen to be $\varepsilon = 10^{-8}$ (default value in PyTorch), then the error constant would be as large as 10^{16} and would make the error estimate essentially not useable in practically relevant regimes.

On a technical level, the key innovative contribution of this work – which allows us to prove convergence rates with full control of the hyperparameters – is to establish and employ new upper bounds for the *inverse moments* of the i -th component of the second moment process $(v_n)_{n \in \mathbb{N}_0}$ in (4) in the [RMSprop](#) method at time $n \in \mathbb{N}$ after the second moment process has become sufficiently large, that is, restricted to the event $\{\sigma^{(i)} \leq n\}$ where $\sigma^{(i)} = \inf\{n \in \mathbb{N}: v_n^{(i)} \geq \iota\}$ for an arbitrary given real constant $\iota \in (0, \infty)$. Such **inverse moment estimates** are established in [Section 4](#) (see [Lemma 4.4](#) and [Lemma 4.5](#) in [Section 4](#)) and are then used extensively in the proof of [Theorem 7.1](#) in [Section 7](#) and, thereby, also in the proof of [Theorem 1.1](#) (see the end of [Section 7](#)).

Such *inverse moment bounds* for $(v_n)_{n \in \mathbb{N}_0}$ cannot in general hold on the whole probability space: before a component of v_n has reached a *non-degenerate level* (that is, has become greater than or equal to some sufficiently large real threshold $\iota \in (0, \infty)$), it may vanish and its inverse moments may explode and fail to

exist. We therefore introduce the random, coordinatewise startup time $\sigma^{(i)}$ at which this level is first reached (at which the i -th component of the second moment process $(v_n^{(i)})_{n \in \mathbb{N}_0}$ has become greater than or equal to a predetermined real threshold ι).

On the complement of the event $\{\sigma^{(i)} \leq n\}$ we use elementary estimates for $F(\theta_n)\mathbb{1}_{\{\tau \geq n\}}$ and establish that the probability of the event $(\Omega \setminus \{\sigma^{(i)} \leq n\}) \cap \{\tau \geq n\} = \{\tau \geq n, \sigma^{(i)} > n\}$ is exponentially small in n (see [Proposition 6.3](#) in [Section 6](#) below).

Performing this early-late decomposition separately for every coordinate and at every iteration produces a single recursion that is valid from the first step on. To the best of our knowledge, this combination of *post-startup inverse moment estimates* with a *different pre-startup argument* is a new *proof mechanism* in the analysis of stochastic gradient methods.

1.5. Overview and structure of this work. The remainder of the article is organized as follows.

- In [Proposition 2.1](#) in [Section 2](#) we derive under the abstract assumptions in [items \(i\)–\(iii\)](#) a one-step recursion for the scalar quantity $\psi_n = \mathbb{E}[F(\theta_n)\mathbb{1}_{\{\tau \geq n\}}]$ (expectation of the τ -stopped evaluation of the objective function $F: \mathbb{R}^d \rightarrow [0, \infty)$ at the [RMSprop](#) optimization process θ_n after n steps) for $n \in \mathbb{N}$. In [Proposition 2.1](#) the events $G_{n,i} \in \mathcal{F}_{n-1}$, $n \in \mathbb{N}$, $i \in \{1, \dots, d\}$, are general predictable events but in our later application of [Proposition 2.1](#) the event $G_{n,i}$ will coincide with the event $\{\sigma^{(i)} < n\}$ where $\sigma^{(i)} = \inf\{n \in \mathbb{N}: v_n^{(i)} \geq (\kappa_0)^{-1}2c\}$ is the stopping time describing the first time where the i -th component of the second moment process $(v_n^{(i)})_{n \in \mathbb{N}_0}$ in the [RMSprop](#) method is sufficiently large (greater than or equal to the constant $2c(\kappa_0)^{-1}$).
- In [Section 3](#) we show that suitable inverse moment bounds for the i -th component of the second moment process $(v_n^{(i)})_{n \in \mathbb{N}_0}$ in the [RMSprop](#) method (see (35) in [Proposition 3.1](#) in [Section 3](#)) are sufficient to ensure that the property in [item \(iii\)](#) in [Proposition 2.1](#) is satisfied (estimates for the *covariance correction*).
- In [Section 4](#) we then establish such *inverse moment bounds* restricted to the events $G_{n,i}$, $n \in \mathbb{N}$, $i \in \{1, \dots, d\}$ (on the complement $\Omega \setminus G_{n,i}$ of the event $G_{n,i}$ the inverse moment bounds do in general *not* hold).
- In [Section 5](#) we provide sufficient conditions to ensure that the property in [item \(ii\)](#) of [Proposition 2.1](#) is satisfied (estimates for the *quadratic Taylor remainder*).
- In [Section 6](#) we show that the probability of the event $(\Omega \setminus G_{n,i}) \cap \{\tau \geq n\}$ is exponentially small in n .
- In [Section 7](#) we combine [Proposition 2.1](#) from [Section 2](#) with the estimates from [Sections 3–6](#) to establish the key error estimate in [Theorem 7.1](#). In addition, in [Section 7](#) we also apply [Theorem 7.1](#) to deduce [Theorem 1.1](#) in this introductory section. In the proof of [Theorem 7.1](#) we apply [Proposition 2.1](#) with the coordinate events $G_{n,i}$. The contributions on their complements are controlled by the estimates from [Section 6](#), while the findings of [Sections 3–5](#) ensure that the assumptions in [items \(ii\) and \(iii\)](#) of [Proposition 2.1](#) are satisfied.
- In [Section 8](#) we illustrate [Theorem 1.1](#) by means of several examples. In particular, in [Section 8](#) we show how [Theorem 1.1](#) can be used to prove [Theorem 1.2](#).

1.6. Use of large language models. The key ideas of the statements and the proofs of the main results of this work are due to the authors. GPT 5.5 and GPT 5.6 have been substantially employed to formulate and work out the statements and the proofs in L^AT_EX according to the strategies of the authors. GPT 5.5 and GPT 5.6 have also supported us in creating the review in [Subsection 1.4](#) and the explanatory texts in the remainder of this work. The authors take full responsibility for each of the statements/arguments/sentences made in this work.

2. THE ABSTRACT RECURSION ESTIMATE IN [PROPOSITION 2.1](#)

This section derives an abstract one-step recursion for the stopped Lyapunov values $F(\theta_n)$. Starting from the smoothness inequality, we separate the coercive drift, the covariance correction caused by the dependence

of v_n on the current innovation, and the quadratic Taylor remainder. The resulting scalar recursion reduces the proof of the main result to estimates for these three contributions.

Proposition 2.1. *Let $L_f, c_v, c_{\text{Loj}}, C_{\text{Loj}} \in (0, \infty)$, $n_0 \in \mathbb{N}$, let $(C_X(n))_{n>n_0}$ and $(C_\beta(n))_{n>n_0}$ be $(0, \infty)$ -valued sequences and let $V \subseteq \mathbb{R}^d$ be measurable. Define*

$$\tau := \inf\{n \in \mathbb{N} \cap [n_0, \infty) : \theta_n \notin V\}. \quad (18)$$

For every $n \in \mathbb{N} \cap (n_0, \infty)$, $i \in \{1, \dots, d\}$ let $G_{n,i} \in \mathcal{F}_{n-1}$. Suppose that $F: \mathbb{R}^d \rightarrow [0, \infty)$ has an L_f -Lipschitz continuous derivative and assume for all $\theta \in V$ that

$$f_X(\theta) = -\nabla F(\theta) \quad \text{and} \quad c_{\text{Loj}}|f_X(\theta)|^2 \leq F(\theta) \leq C_{\text{Loj}}|f_X(\theta)|^2. \quad (19)$$

Assume moreover that all quantities appearing below are well-defined and integrable to the powers used there on the corresponding stopped events. If $\varepsilon = 0$, inverse factors and conditional covariances appearing together with $\mathbb{1}_{G_{n,i}}$ are evaluated on $\{\tau \geq n\} \cap G_{n,i}$ and extended by zero to its complement. Assume that, for every $n \in \mathbb{N} \cap (n_0, \infty)$, the following conditions¹ hold:

- (i) $\mathbb{E}[\mathbb{1}_{\{\tau \geq n\}} \sum_{i=1}^d \mathbb{1}_{G_{n,i}} (\sqrt{v_n^{(i)}} + \varepsilon)^{-1} |f_X^{(i)}(\theta_{n-1})|^2] \geq c_v \mathbb{E}[\mathbb{1}_{\{\tau \geq n\}} \sum_{i=1}^d \mathbb{1}_{G_{n,i}} |f_X^{(i)}(\theta_{n-1})|^2],$
- (ii) $\mathbb{E}[\mathbb{1}_{\{\tau \geq n\}} \sum_{i=1}^d \mathbb{1}_{G_{n,i}} (\sqrt{v_n^{(i)}} + \varepsilon)^{-2} X^{(i)}(\theta_{n-1}, U_n)^2] \leq C_X(n),$ and
- (iii) $\mathbb{E}[\mathbb{1}_{\{\tau \geq n\}} \sum_{i=1}^d \mathbb{1}_{G_{n,i}} \text{cov}((\sqrt{v_n^{(i)}} + \varepsilon)^{-1}, X^{(i)}(\theta_{n-1}, U_n) | \mathcal{F}_{n-1})^2]^{1/2} \leq C_\beta(n).$

For every $n \in \mathbb{N} \cap (n_0, \infty)$, set

$$\psi_n = \mathbb{E}[\mathbb{1}_{\{\tau \geq n\}} F(\theta_n)] \quad \text{and} \quad \tilde{\psi}_n = \mathbb{E}[\mathbb{1}_{\{\tau \geq n\}} F(\theta_{n-1})]. \quad (20)$$

Then one has for every $n \in \mathbb{N} \cap (n_0, \infty)$ that

$$\begin{aligned} \psi_n &\leq \left(1 - \frac{c_v}{C_{\text{Loj}}} \gamma_n\right) \tilde{\psi}_n + \frac{C_\beta(n)}{c_{\text{Loj}}^{1/2}} \gamma_n \tilde{\psi}_n^{1/2} + \frac{L_f C_X(n)}{2} \gamma_n^2 \\ &\quad + c_v \gamma_n \mathbb{E}\left[\mathbb{1}_{\{\tau \geq n\}} \sum_{i=1}^d \mathbb{1}_{G_{n,i}^c} |f_X^{(i)}(\theta_{n-1})|^2\right] + \gamma_n \mathbb{E}\left[\mathbb{1}_{\{\tau \geq n\}} \sum_{i=1}^d \mathbb{1}_{G_{n,i}^c} |f_X^{(i)}(\theta_{n-1})| (\sqrt{v_n^{(i)}} + \varepsilon)^{-1} X^{(i)}(\theta_{n-1}, U_n)\right] \\ &\quad + \frac{L_f \gamma_n^2}{2} \mathbb{E}\left[\mathbb{1}_{\{\tau \geq n\}} \sum_{i=1}^d \mathbb{1}_{G_{n,i}^c} (\sqrt{v_n^{(i)}} + \varepsilon)^{-2} X^{(i)}(\theta_{n-1}, U_n)^2\right]. \end{aligned} \quad (21)$$

In particular, one has for every $n \in \mathbb{N} \cap (n_0, \infty)$ that

$$\begin{aligned} \psi_n &\leq \left(1 - \frac{c_v}{2C_{\text{Loj}}} \gamma_n\right) \tilde{\psi}_n + \frac{C_\beta(n)^2 C_{\text{Loj}}}{2c_v C_{\text{Loj}}} \gamma_n + \frac{L_f C_X(n)}{2} \gamma_n^2 \\ &\quad + c_v \gamma_n \mathbb{E}\left[\mathbb{1}_{\{\tau \geq n\}} \sum_{i=1}^d \mathbb{1}_{G_{n,i}^c} |f_X^{(i)}(\theta_{n-1})|^2\right] + \gamma_n \mathbb{E}\left[\mathbb{1}_{\{\tau \geq n\}} \sum_{i=1}^d \mathbb{1}_{G_{n,i}^c} |f_X^{(i)}(\theta_{n-1})| (\sqrt{v_n^{(i)}} + \varepsilon)^{-1} X^{(i)}(\theta_{n-1}, U_n)\right] \\ &\quad + \frac{L_f \gamma_n^2}{2} \mathbb{E}\left[\mathbb{1}_{\{\tau \geq n\}} \sum_{i=1}^d \mathbb{1}_{G_{n,i}^c} (\sqrt{v_n^{(i)}} + \varepsilon)^{-2} X^{(i)}(\theta_{n-1}, U_n)^2\right]. \end{aligned} \quad (22)$$

Proof. Fix $n > n_0$. Since τ is the first exit time from V after time n_0 , the event $\{\tau \geq n\}$ implies $\theta_{n-1} \in V$. Since ∇F is L_f -Lipschitz continuous and $f_X = -\nabla F$ on V we get that for all $x \in V$ and $y \in \mathbb{R}^d$

$$F(y) \leq F(x) - \langle f_X(x), y - x \rangle + \frac{L_f}{2} |y - x|^2. \quad (23)$$

Using that

$$\theta_n = \theta_{n-1} + \gamma_n (\sqrt{v_n} + \varepsilon)^{\circ(-1)} \circ X(\theta_{n-1}, U_n), \quad (24)$$

¹Note that for every probability space $(\Omega, \mathcal{F}, \mathbb{P})$, every sigma-algebra $\mathcal{G} \subseteq \mathcal{F}$ on Ω and all square integrable random variables $X: \Omega \rightarrow \mathbb{R}$ and $Y: \Omega \rightarrow \mathbb{R}$ it holds $\mathbb{P}$ -a.s. that $\text{cov}(X, Y | \mathcal{G}) = \mathbb{E}[(X - \mathbb{E}[X | \mathcal{G}])(Y - \mathbb{E}[Y | \mathcal{G}] | \mathcal{G})]$.

Combining (23) and (24), we conclude that on the event $\{\tau \geq n\}$

$$\begin{aligned} F(\theta_n) &\leq F(\theta_{n-1}) - \gamma_n \sum_{i=1}^d (\sqrt{v_n^{(i)}} + \varepsilon)^{-1} f_X^{(i)}(\theta_{n-1}) X^{(i)}(\theta_{n-1}, U_n) \\ &\quad + \frac{L_f \gamma_n^2}{2} \sum_{i=1}^d (\sqrt{v_n^{(i)}} + \varepsilon)^{-2} X^{(i)}(\theta_{n-1}, U_n)^2. \end{aligned} \quad (25)$$

The events $G_{n,i}$ and $\{\tau \geq n\}$ are $\mathcal{F}_{n-1}$ -measurable. Moreover, $\mathbb{E}[X^{(i)}(\theta_{n-1}, U_n) | \mathcal{F}_{n-1}] = f_X^{(i)}(\theta_{n-1})$. Hence, multiplying (25) by $\mathbb{1}_{\{\tau \geq n\}}$, using the conditional covariance decomposition on $G_{n,i}$, and taking expectations, we arrive at

$$\begin{aligned} \psi_n &\leq \tilde{\psi}_n - \gamma_n \mathbb{E} \left[\mathbb{1}_{\{\tau \geq n\}} \sum_{i=1}^d \mathbb{1}_{G_{n,i}} (\sqrt{v_n^{(i)}} + \varepsilon)^{-1} |f_X^{(i)}(\theta_{n-1})|^2 \right] \\ &\quad - \gamma_n \mathbb{E} \left[\mathbb{1}_{\{\tau \geq n\}} \sum_{i=1}^d \mathbb{1}_{G_{n,i}} f_X^{(i)}(\theta_{n-1}) \text{cov}((\sqrt{v_n^{(i)}} + \varepsilon)^{-1}, X^{(i)}(\theta_{n-1}, U_n) | \mathcal{F}_{n-1}) \right] \\ &\quad + \gamma_n \mathbb{E} \left[\mathbb{1}_{\{\tau \geq n\}} \sum_{i=1}^d \mathbb{1}_{G_{n,i}^c} |f_X^{(i)}(\theta_{n-1})| (\sqrt{v_n^{(i)}} + \varepsilon)^{-1} X^{(i)}(\theta_{n-1}, U_n) \right] \\ &\quad + \frac{L_f \gamma_n^2}{2} \mathbb{E} \left[\mathbb{1}_{\{\tau \geq n\}} \sum_{i=1}^d \mathbb{1}_{G_{n,i}} (\sqrt{v_n^{(i)}} + \varepsilon)^{-2} X^{(i)}(\theta_{n-1}, U_n)^2 \right] \\ &\quad + \frac{L_f \gamma_n^2}{2} \mathbb{E} \left[\mathbb{1}_{\{\tau \geq n\}} \sum_{i=1}^d \mathbb{1}_{G_{n,i}^c} (\sqrt{v_n^{(i)}} + \varepsilon)^{-2} X^{(i)}(\theta_{n-1}, U_n)^2 \right]. \end{aligned} \quad (26)$$

Next, we estimate the drift, covariance, and quadratic terms on the right-hand side. By assumption (i) and the upper Łojasiewicz bound, we have

$$\begin{aligned} &\mathbb{E} \left[\mathbb{1}_{\{\tau \geq n\}} \sum_{i=1}^d \mathbb{1}_{G_{n,i}} (\sqrt{v_n^{(i)}} + \varepsilon)^{-1} |f_X^{(i)}(\theta_{n-1})|^2 \right] \\ &\geq c_v \mathbb{E} \left[\mathbb{1}_{\{\tau \geq n\}} \sum_{i=1}^d \mathbb{1}_{G_{n,i}} |f_X^{(i)}(\theta_{n-1})|^2 \right] \\ &\geq \frac{c_v}{C_{\text{Loj}}} \tilde{\psi}_n - c_v \mathbb{E} \left[\mathbb{1}_{\{\tau \geq n\}} \sum_{i=1}^d \mathbb{1}_{G_{n,i}^c} |f_X^{(i)}(\theta_{n-1})|^2 \right]. \end{aligned} \quad (27)$$

Next, we estimate the covariance term. As a consequence of the Cauchy–Schwarz inequality, we have

$$\begin{aligned} &\left| \mathbb{E} \left[\mathbb{1}_{\{\tau \geq n\}} \sum_{i=1}^d \mathbb{1}_{G_{n,i}} f_X^{(i)}(\theta_{n-1}) \text{cov}((\sqrt{v_n^{(i)}} + \varepsilon)^{-1}, X^{(i)}(\theta_{n-1}, U_n) | \mathcal{F}_{n-1}) \right] \right| \\ &\leq \mathbb{E} \left[\mathbb{1}_{\{\tau \geq n\}} |f_X(\theta_{n-1})|^2 \right]^{1/2} \mathbb{E} \left[\mathbb{1}_{\{\tau \geq n\}} \sum_{i=1}^d \mathbb{1}_{G_{n,i}} \text{cov}((\sqrt{v_n^{(i)}} + \varepsilon)^{-1}, X^{(i)}(\theta_{n-1}, U_n) | \mathcal{F}_{n-1})^2 \right]^{1/2}. \end{aligned} \quad (28)$$

By assumption (iii), the second factor is bounded by $C_\beta(n)$. Moreover, on $\{\tau \geq n\}$ the lower Łojasiewicz bound yields

$$|f_X(\theta_{n-1})|^2 \leq c_{\text{Loj}}^{-1} F(\theta_{n-1}), \quad (29)$$

so that (20) and (29) give

$$\mathbb{E} \left[\mathbb{1}_{\{\tau \geq n\}} |f_X(\theta_{n-1})|^2 \right] \leq c_{\text{Loj}}^{-1} \tilde{\psi}_n. \quad (30)$$

Consequently, we get with (28) and (30) that

$$\mathbb{E} \left[\mathbb{1}_{\{\tau \geq n\}} \sum_{i=1}^d \mathbb{1}_{G_{n,i}} f_X^{(i)}(\theta_{n-1}) \text{cov}((\sqrt{v_n^{(i)}} + \varepsilon)^{-1}, X^{(i)}(\theta_{n-1}, U_n) | \mathcal{F}_{n-1}) \right] \geq -\frac{C_\beta(n)}{c_{\text{Loj}}^{1/2}} \tilde{\psi}_n^{1/2}. \quad (31)$$

By assumption (ii),

$$\mathbb{E}\left[\mathbb{1}_{\{\tau \geq n\}} \sum_{i=1}^d \mathbb{1}_{G_{n,i}} (\sqrt{v_n^{(i)}} + \varepsilon)^{-2} X^{(i)}(\theta_{n-1}, U_n)^2\right] \leq C_X(n). \quad (32)$$

Combining (26) with (27), (31), and (32) proves (21). For the second estimate, we use Young's inequality in the form

$$\frac{C_\beta(n)}{c_{\text{Loj}}^{1/2}} \tilde{\psi}_n^{1/2} \leq \frac{c_v}{2C_{\text{Loj}}} \tilde{\psi}_n + \frac{C_\beta(n)^2 C_{\text{Loj}}}{2c_v c_{\text{Loj}}}, \quad (33)$$

which, together with (21) and (33), proves (22). $\square$

3. ANALYSIS OF THE PROPERTY IN ITEM (III) OF PROPOSITION 2.1

This section controls the covariance correction appearing in item (iii) of Proposition 2.1. A conditional Lipschitz argument bounds each coordinatewise covariance in terms of a conditional third moment of the innovation and a negative third moment of v_{n-1} . An auxiliary estimate for the resulting β -dependent quotient then distinguishes the initial n^{-1} -regime from the asymptotic $1 - \beta$ -regime.

Proposition 3.1. *Let $\beta \in [1/2, 1)$, $\varepsilon \in [0, \infty)$, $i \in \{1, 2, \dots, d\}$, $n \in \mathbb{N} \setminus \{1\}$, let $A_n \in \mathcal{F}_{n-1}$, and let $C_X^{(3)}, C_v^{(-3)} \in (0, \infty)$. Assume that*

$$\mathbb{E}[|X^{(i)}(\theta_{n-1}, U_n)|^3 | \mathcal{F}_{n-1}] \leq C_X^{(3)} \quad \text{on } A_n \quad (34)$$

and

$$\mathbb{E}[\mathbb{1}_{A_n} (v_{n-1}^{(i)})^{-3}] \leq C_v^{(-3)}. \quad (35)$$

The negative-moment assumption implies that $v_{n-1}^{(i)} > 0$ almost surely on A_n . If $\varepsilon = 0$, the conditional covariances in the conclusion are therefore evaluated on A_n and extended by zero to $\Omega \setminus A_n$. Then one has

$$\mathbb{E}\left[\mathbb{1}_{A_n} \text{cov}((\sqrt{v_n^{(i)}} + \varepsilon)^{-1}, X^{(i)}(\theta_{n-1}, U_n) | \mathcal{F}_{n-1})^2\right]^{1/2} \leq 4C_X^{(3)} (C_v^{(-3)})^{1/2} \frac{1 - \beta}{1 - \beta^{n-1}}. \quad (36)$$

Proof. Set

$$Y_n^{(i)} = \frac{\beta(1 - \beta^{n-1})}{1 - \beta^n} v_{n-1}^{(i)} = \kappa_n v_{n-1}^{(i)} \quad \text{and} \quad Z_n = \frac{1 - \beta}{1 - \beta^n}. \quad (37)$$

Then

$$v_n^{(i)} = Y_n^{(i)} + Z_n X^{(i)}(\theta_{n-1}, U_n)^2. \quad (38)$$

The negative-moment assumption gives $v_{n-1}^{(i)} > 0$ almost surely on A_n , and hence $Y_n^{(i)} > 0$ there. In what follows, all conditional identities involving inverse factors are understood on A_n . The function

$$h : (0, \infty) \rightarrow (0, \infty), \quad y \mapsto h(y) = \frac{1}{\sqrt{y} + \varepsilon} \quad (39)$$

is for every $y_0 \in (0, \infty)$ Lipschitz continuous on $[y_0, \infty)$ with constant $\frac{1}{2}y_0^{-3/2}$. Using that $Y_n^{(i)}$ is $\mathcal{F}_{n-1}$ -measurable, we conclude on A_n that

$$\begin{aligned}
& |\text{cov}((\sqrt{v_n^{(i)}} + \varepsilon)^{-1}, X^{(i)}(\theta_{n-1}, U_n) | \mathcal{F}_{n-1})| \\
&= |\text{cov}(h(Y_n^{(i)} + Z_n X^{(i)}(\theta_{n-1}, U_n)^2), X^{(i)}(\theta_{n-1}, U_n) | \mathcal{F}_{n-1})| \\
&= \left| \mathbb{E} \left[(h(Y_n^{(i)} + Z_n X^{(i)}(\theta_{n-1}, U_n)^2) - h(Y_n^{(i)})) (X^{(i)}(\theta_{n-1}, U_n) - f_X^{(i)}(\theta_{n-1})) \mid \mathcal{F}_{n-1} \right] \right| \\
&\leq \frac{Z_n}{2(Y_n^{(i)})^{3/2}} \mathbb{E} \left[X^{(i)}(\theta_{n-1}, U_n)^2 | X^{(i)}(\theta_{n-1}, U_n) - f_X^{(i)}(\theta_{n-1}) \mid \mathcal{F}_{n-1} \right] \\
&\leq \frac{Z_n}{2(Y_n^{(i)})^{3/2}} \left(\mathbb{E}[|X^{(i)}(\theta_{n-1}, U_n)|^3 | \mathcal{F}_{n-1}] + \mathbb{E}[|X^{(i)}(\theta_{n-1}, U_n)|^2 | \mathcal{F}_{n-1}] \mathbb{E}[|X^{(i)}(\theta_{n-1}, U_n)| | \mathcal{F}_{n-1}] \right) \\
&\leq \frac{Z_n}{(Y_n^{(i)})^{3/2}} \mathbb{E}[|X^{(i)}(\theta_{n-1}, U_n)|^3 | \mathcal{F}_{n-1}].
\end{aligned} \tag{40}$$

On A_n , the assumption on the third moment therefore implies

$$|\text{cov}((\sqrt{v_n^{(i)}} + \varepsilon)^{-1}, X^{(i)}(\theta_{n-1}, U_n) | \mathcal{F}_{n-1})|^2 \leq Z_n^2 (C_X^{(3)})^2 (Y_n^{(i)})^{-3}. \tag{41}$$

Multiplying by $\mathbb{1}_{A_n}$ and taking expectations yields

$$\mathbb{E} \left[\mathbb{1}_{A_n} \text{cov}((\sqrt{v_n^{(i)}} + \varepsilon)^{-1}, X^{(i)}(\theta_{n-1}, U_n) | \mathcal{F}_{n-1})^2 \right] \leq Z_n^2 (C_X^{(3)})^2 \mathbb{E}[\mathbb{1}_{A_n} (Y_n^{(i)})^{-3}]. \tag{42}$$

Since $Y_n^{(i)} = \kappa_n v_{n-1}^{(i)}$, we obtain

$$\mathbb{E}[\mathbb{1}_{A_n} (Y_n^{(i)})^{-3}] = \left(\frac{1 - \beta^n}{\beta(1 - \beta^{n-1})} \right)^3 \mathbb{E}[\mathbb{1}_{A_n} (v_{n-1}^{(i)})^{-3}] \leq \left(\frac{1 - \beta^n}{\beta(1 - \beta^{n-1})} \right)^3 C_v^{(-3)}. \tag{43}$$

Hence, combining (42) and (43) and using the definition of Z_n ,

$$\begin{aligned}
\mathbb{E} \left[\mathbb{1}_{A_n} \text{cov}((\sqrt{v_n^{(i)}} + \varepsilon)^{-1}, X^{(i)}(\theta_{n-1}, U_n) | \mathcal{F}_{n-1})^2 \right] &\leq (C_X^{(3)})^2 C_v^{(-3)} \left(\frac{1 - \beta}{1 - \beta^n} \right)^2 \left(\frac{1 - \beta^n}{\beta(1 - \beta^{n-1})} \right)^3 \\
&= (C_X^{(3)})^2 C_v^{(-3)} \beta^{-3} \frac{(1 - \beta)^2 (1 - \beta^n)}{(1 - \beta^{n-1})^3}.
\end{aligned} \tag{44}$$

By Lemma 4.3, we have

$$\frac{1 - \beta^n}{n} \leq \frac{1 - \beta^{n-1}}{n-1}, \quad \text{so that} \quad \frac{1 - \beta^n}{1 - \beta^{n-1}} \leq \frac{n}{n-1}. \tag{45}$$

we infer, using $\beta \in [\frac{1}{2}, 1)$ and $n \geq 2$, that

$$\beta^{-3} \frac{1 - \beta^n}{1 - \beta^{n-1}} \leq 8 \cdot 2 = 16. \tag{46}$$

Taking square roots, we finish the proof with the conclusion that

$$\mathbb{E} \left[\mathbb{1}_{A_n} \text{cov}((\sqrt{v_n^{(i)}} + \varepsilon)^{-1}, X^{(i)}(\theta_{n-1}, U_n) | \mathcal{F}_{n-1})^2 \right]^{1/2} \leq 4 C_X^{(3)} (C_v^{(-3)})^{1/2} \frac{1 - \beta}{1 - \beta^{n-1}}. \tag{47}$$

□

The latter proposition incorporates an n -dependent term $\frac{1-\beta}{1-\beta^{n-1}}$. This is analysed in the next lemma.

Lemma 3.2. *Let $n \in \mathbb{N}$, $\beta \in [0, 1)$. Then one has that*

$$\frac{1 - \beta}{1 - \beta^n} \leq \begin{cases} \frac{2}{n}, & \text{if } n \leq (1 - \beta)^{-1} \\ \frac{1-\beta}{1-\beta^{n-1}}, & \text{if } n \geq (1 - \beta)^{-1} \end{cases} \tag{48}$$

Proof. Set $x = 1 - \beta \in (0, 1]$. Then

$$\frac{1 - \beta}{1 - \beta^n} = \frac{x}{1 - (1 - x)^n}. \quad (49)$$

Assume first that $n \leq (1 - \beta)^{-1}$, that is, $nx \leq 1$. Since $(1 - x)^n \leq e^{-nx}$ and $1 - e^{-y} \geq y/2$ for every $y \in [0, 1]$, we obtain

$$1 - (1 - x)^n \geq 1 - e^{-nx} \geq \frac{nx}{2}, \quad \text{which yields} \quad \frac{1 - \beta}{1 - \beta^n} = \frac{x}{1 - (1 - x)^n} \leq \frac{2}{n}. \quad (50)$$

This proves the first assertion.

Now assume that $n \geq (1 - \beta)^{-1}$, that is, $nx \geq 1$. Then

$$\beta^n = (1 - x)^n \leq e^{-nx} \leq e^{-1} \quad \text{and, hence,} \quad \frac{1 - \beta}{1 - \beta^n} \leq \frac{1 - \beta}{1 - e^{-1}}. \quad (51)$$

This proves the second assertion. $\square$

4. INVSEIVE MOMENT ESTIMATES FOR THE SECOND MOMENT PROCESS $(v_n)_{n \in \mathbb{N}_0}$ IN THE RMSPPROP METHOD

This section establishes the negative moment estimates needed for the covariance and quadratic error bounds. We introduce the (c, q) -regularity condition, derive conditional Laplace-transform estimates for the stopped conditioner, and convert these estimates into bounds for arbitrary negative moments. Since such bounds need not hold initially, the final result is formulated on the event that the corresponding coordinatewise startup time has already occurred.

Definition 4.1. Let $c, q \in (0, \infty)$. We say that an innovation (X, U) is (c, q) -regular on a set $\mathcal{V} \subseteq \mathbb{R}^d$ if and only if for all $\theta \in \mathcal{V}$, $i \in \{1, \dots, d\}$, $\lambda \in [0, \infty)$ we have that

$$-\ln \mathbb{E}[e^{-\lambda X^{(i)}(\theta, U)^2}] \geq (c\lambda) \wedge q. \quad (52)$$

First we will deduce that (c, q) -regular innovations do allow us to estimate the exponential moments of v_n .

Proposition 4.2. Let (X, U) be a (c, q) -regular innovation on a measurable set $\mathcal{V}$ and let (θ_n) be an *RMSpprop* algorithm with arbitrary parameter tuple $(\beta, \varepsilon, (\gamma_n), \theta_0)$ as in the introduction. Let $n_0 \in \mathbb{N}_0$ and let τ be the stopping time given by

$$\tau = \inf\{n \geq n_0 : \theta_n \notin \mathcal{V}\}. \quad (53)$$

One has for every $n \in \mathbb{N}_0 \cap [n_0, \infty)$, $i \in \{1, \dots, d\}$, $\lambda \in [0, \infty)$ that

$$-\ln \mathbb{E}[e^{-\lambda(v_n^{(i)} + \mathbf{1}_{\{n \leq \tau\}} c \infty)} | \mathcal{F}_{n_0}] \geq \sum_{k=n_0+1}^n \left(c \frac{1 - \beta}{1 - \beta^n} \beta^{n-k} \lambda \right) \wedge q + \frac{1 - \beta^{n_0}}{1 - \beta^n} \beta^{n-n_0} \lambda v_{n_0}^{(i)}. \quad (54)$$

In the case $n = n_0 = 0$, the last term on the right-hand side is understood as zero.

Proof. Fix $i \in \{1, \dots, d\}$, $\lambda \in [0, \infty)$. The statement is trivial for $n = n_0$. We thus fix $n \in \mathbb{N}$ with $n > n_0$ and note that

$$\begin{aligned} v_n^{(i)} &= \frac{1 - \beta}{1 - \beta^n} \sum_{k=1}^n \beta^{n-k} X^{(i)}(\theta_{k-1}, U_k)^2 \\ &= \sum_{k=n_0+1}^n \frac{1 - \beta}{1 - \beta^n} \beta^{n-k} X^{(i)}(\theta_{k-1}, U_k)^2 + v_{n_0}^{(i)} \frac{1 - \beta^{n_0}}{1 - \beta^n} \beta^{n-n_0}. \end{aligned} \quad (55)$$

We let for $k = n_0 + 1, \dots, n$

$$Z_k := \begin{cases} \frac{1 - \beta}{1 - \beta^n} \beta^{n-k} X^{(i)}(\theta_{k-1}, U_k)^2, & \text{if } k \leq \tau, \\ \infty, & \text{else.} \end{cases} \quad (56)$$

Then one has

$$v_n^{(i)} + \mathbb{1}_{\{n \leq \tau\}^c} \infty = \sum_{k=n_0+1}^n Z_k + v_{n_0}^{(i)} \frac{1-\beta^{n_0}}{1-\beta^n} \beta^{n-n_0}. \quad (57)$$

Moreover,

$$\begin{aligned} \mathbb{E}[e^{-\lambda \sum_{k=n_0+1}^n Z_k} | \mathcal{F}_{n-1}] &= e^{-\lambda \sum_{k=n_0+1}^{n-1} Z_k} \mathbb{1}_{\{n \leq \tau\}} \mathbb{E}\left[\exp\left\{-\lambda \frac{1-\beta}{1-\beta^n} X^{(i)}(\theta_{n-1}, U_n)^2\right\} \middle| \mathcal{F}_{n-1}\right] \\ &\leq e^{-\lambda \sum_{k=n_0+1}^{n-1} Z_k} \exp\left\{-\left(c \frac{1-\beta}{1-\beta^n} \lambda\right) \wedge q\right\}. \end{aligned} \quad (58)$$

Iterating yields that

$$\mathbb{E}[e^{-\lambda \sum_{k=n_0+1}^n Z_k} | \mathcal{F}_{n_0}] \leq \exp\left\{-\sum_{k=n_0+1}^n \left(c \frac{1-\beta}{1-\beta^n} \beta^{n-k} \lambda\right) \wedge q\right\}. \quad (59)$$

Consequently, one has that

$$\begin{aligned} \mathbb{E}[e^{-\lambda(v_n^{(i)} + \mathbb{1}_{\{n \leq \tau\}^c} \infty)} | \mathcal{F}_{n_0}] &= e^{-\lambda v_{n_0}^{(i)} \frac{1-\beta^{n_0}}{1-\beta^n} \beta^{n-n_0}} \mathbb{E}[e^{-\lambda \sum_{k=n_0+1}^n Z_k} | \mathcal{F}_{n_0}] \\ &\leq \exp\left\{-\sum_{k=n_0+1}^n \left(c \frac{1-\beta}{1-\beta^n} \beta^{n-k} \lambda\right) \wedge q - \frac{1-\beta^{n_0}}{1-\beta^n} \beta^{n-n_0} \lambda v_{n_0}^{(i)}\right\}. \end{aligned} \quad (60)$$

This finishes the proof. $\square$

Next, we will deduce a negative moment estimate for random variables satisfying an exponential moment estimate like (54). The moment estimate will make use of the following technical result.

Lemma 4.3. *Let $\beta \in [0, 1)$. The function*

$$f : (0, \infty) \rightarrow (0, \infty), \quad f(x) = \frac{1-\beta^x}{x} \quad (61)$$

is decreasing. In particular, for every $n \in \mathbb{N}$,

$$\frac{1-\beta}{1-\beta^n} \geq \frac{1}{n}. \quad (62)$$

Proof. If $\beta = 0$, then $f(x) = x^{-1}$ for every $x > 0$, so monotonicity is clear. Now suppose that $\beta \in (0, 1)$. For $x > 0$, one has

$$f'(x) = \frac{-x\beta^x \ln \beta - 1 + \beta^x}{x^2}. \quad (63)$$

Letting $y = -x \ln \beta > 0$, the numerator becomes

$$(y+1)e^{-y} - 1. \quad (64)$$

Since $e^y > 1+y$ for all $y > 0$, we obtain $(y+1)e^{-y} < 1$. Hence the numerator is strictly negative, so that $f'(x) < 0$ for all $x > 0$. This proves the monotonicity assertion.

For every $n \in \mathbb{N}$, monotonicity implies that

$$\frac{1-\beta^n}{n} = f(n) \leq f(1) = 1-\beta, \quad (65)$$

which implies the stated inequality. $\square$

Lemma 4.4 (Inverse moment estimates). *Let $c, q, \rho \in (0, \infty)$, $p \in [1, \infty)$, $\beta \in (0, 1)$, $n, n_0 \in \mathbb{N}$ satisfy*

$$q + p \ln \beta \geq \rho \quad (66)$$

and $n > n_0$. Let V_n^ be a non-negative random variable which satisfies for all $\lambda \in [0, \infty)$ that*

$$-\ln \mathbb{E}[e^{-\lambda V_n^*}] \geq \sum_{k=n_0+1}^n \left(c \frac{1-\beta}{1-\beta^n} \beta^{n-k} \lambda\right) \wedge q + c \frac{1-\beta^{n_0}}{1-\beta^n} \beta^{n-n_0} \lambda. \quad (67)$$

Then

$$\mathbb{E}[(V_n^*)^{-p}] \leq \frac{1}{c^p} D_{\rho,p}, \quad (68)$$

where

$$D_{\rho,p} := \sum_{r=0}^{\infty} (r+1)^p e^{-\rho r} < \infty. \quad (69)$$

Proof. We can assume without loss of generality that $c = 1$. Note that for every $x \geq 0$

$$x^{-p} = \frac{1}{\Gamma(p)} \int_0^{\infty} e^{-x\lambda} \lambda^{p-1} d\lambda \quad (70)$$

so that by Fubini's theorem

$$\mathbb{E}[(V_n^*)^{-p}] = \frac{1}{\Gamma(p)} \mathbb{E} \left[\int_0^{\infty} e^{-V_n^* \lambda} \lambda^{p-1} d\lambda \right] = \frac{1}{\Gamma(p)} \int_0^{\infty} \mathbb{E}[e^{-\lambda V_n^*}] \lambda^{p-1} d\lambda. \quad (71)$$

To bound the integral we distinguish cases and choose

$$C_1 = \frac{q(1-\beta^n)}{1-\beta} \quad \text{and} \quad C_2 = \frac{q(1-\beta^n)}{(1-\beta)\beta^{n-n_0-1}} = C_1 \beta^{-(n-n_0-1)}. \quad (72)$$

Thus

$$\begin{aligned} \mathbb{E}[(V_n^*)^{-p}] &= \frac{1}{\Gamma(p)} \int_0^{C_1} \mathbb{E}[e^{-\lambda V_n^*}] \lambda^{p-1} d\lambda + \frac{1}{\Gamma(p)} \int_{C_1}^{C_2} \mathbb{E}[e^{-\lambda V_n^*}] \lambda^{p-1} d\lambda \\ &\quad + \frac{1}{\Gamma(p)} \int_{C_2}^{\infty} \mathbb{E}[e^{-\lambda V_n^*}] \lambda^{p-1} d\lambda. \end{aligned} \quad (73)$$

1) First, let us estimate the contribution of $[C_1, C_2)$ by slicing this interval. If

$$C_1 < C_2, \quad (74)$$

equivalently if $n \geq n_0 + 2$, then for $r = 1, \dots, n - n_0 - 1$ define

$$I_r := [\beta^{-(r-1)} C_1, \beta^{-r} C_1). \quad (75)$$

These slices form a disjoint partition of $[C_1, C_2)$. For every $\lambda \in I_r$ one has

$$\beta^{-(r-1)} C_1 \leq \lambda < \beta^{-r} C_1 \quad \text{or, equivalently,} \quad \beta^{-(r-1)} q \leq \frac{1-\beta}{1-\beta^n} \lambda < \beta^{-r} q. \quad (76)$$

Hence, for $k \in \{n_0 + 1, \dots, n\}$, one has

$$\frac{1-\beta}{1-\beta^n} \beta^{n-k} \lambda \begin{cases} < q & \text{if } k \in \{n_0 + 1, \dots, n-r\}, \\ \geq q & \text{if } k \in \{n-r+1, \dots, n\}, \end{cases} \quad (77)$$

so that

$$\begin{aligned} \sum_{k=n_0+1}^n \left(\frac{1-\beta}{1-\beta^n} \beta^{n-k} \lambda \right) \wedge q &= \sum_{k=n_0+1}^{n-r} \frac{1-\beta}{1-\beta^n} \beta^{n-k} \lambda + r q \\ &= \frac{\beta^r - \beta^n}{1-\beta^n} \lambda + r q. \end{aligned} \quad (78)$$

Combining this with (67), we obtain

$$-\ln \mathbb{E}[e^{-\lambda V_n^*}] \geq r q + \lambda \frac{\beta^r - \beta^n}{1-\beta^n} = r q + a_{n,r} \lambda, \quad (79)$$

where

$$a_{n,r} := \frac{\beta^r - \beta^n}{1-\beta^n} = \beta^r \frac{1-\beta^{n-r}}{1-\beta^n}. \quad (80)$$

Therefore, we obtain that

$$\begin{aligned}
J_r &:= \frac{1}{\Gamma(p)} \int_{I_r} \mathbb{E}[e^{-\lambda V_n^*}] \lambda^{p-1} d\lambda \leq \frac{e^{-rq}}{\Gamma(p)} \int_{\beta^{-(r-1)C_1}}^{\beta^{-r}C_1} e^{-a_{n,r}\lambda} \lambda^{p-1} d\lambda \\
&= \frac{e^{-rq}}{\Gamma(p)} a_{n,r}^{-p} \int_{a_{n,r}\beta^{-(r-1)C_1}}^{a_{n,r}\beta^{-r}C_1} e^{-u} u^{p-1} du \\
&\leq \frac{e^{-rq}}{\Gamma(p)} a_{n,r}^{-p} \int_0^\infty e^{-u} u^{p-1} du = e^{-rq} a_{n,r}^{-p} \\
&= e^{-r(q+p \ln \beta)} \left(\frac{1 - \beta^n}{1 - \beta^{n-r}} \right)^p.
\end{aligned} \tag{81}$$

Using [Lemma 4.3](#) and that $n - r \geq 1$, we get that

$$\frac{1 - \beta^n}{1 - \beta^{n-r}} \leq \frac{n}{n-r} \leq r+1. \tag{82}$$

Consequently, we can conclude with assumption (66) that

$$J_r \leq (r+1)^p e^{-r(q+p \ln \beta)} \leq (r+1)^p e^{-\rho r}. \tag{83}$$

Hence the whole contribution of the middle interval is bounded by

$$\frac{1}{\Gamma(p)} \int_{C_1}^{C_2} \mathbb{E}[e^{-\lambda V_n^*}] \lambda^{p-1} d\lambda = \sum_{r=1}^{n-n_0-1} J_r \leq \sum_{r=1}^{n-n_0-1} (r+1)^p e^{-\rho r}, \tag{84}$$

while for $n = n_0 + 1$ the interval $[C_1, C_2)$ is empty.

2) Next, let $\lambda \geq C_2$. Then all terms in the sum are capped at q , so

$$-\ln \mathbb{E}[e^{-\lambda V_n^*}] \geq (n - n_0)q + \frac{1 - \beta^{n_0}}{1 - \beta^n} \beta^{n-n_0} \lambda. \tag{85}$$

Therefore,

$$\begin{aligned}
\frac{1}{\Gamma(p)} \int_{C_2}^\infty \mathbb{E}[e^{-\lambda V_n^*}] \lambda^{p-1} d\lambda &\leq \frac{e^{-(n-n_0)q}}{\Gamma(p)} \int_{C_2}^\infty \exp\left(-\frac{1 - \beta^{n_0}}{1 - \beta^n} \beta^{n-n_0} \lambda\right) \lambda^{p-1} d\lambda \\
&\leq \frac{e^{-(n-n_0)q}}{\Gamma(p)} \int_0^\infty \exp\left(-\frac{1 - \beta^{n_0}}{1 - \beta^n} \beta^{n-n_0} \lambda\right) \lambda^{p-1} d\lambda \\
&= \left(\frac{1 - \beta^n}{(1 - \beta^{n_0}) \beta^{n-n_0}} \right)^p e^{-(n-n_0)q},
\end{aligned} \tag{86}$$

where we have used (70) again. Applying [Lemma 4.3](#), we obtain

$$\frac{1 - \beta^n}{n} \leq \frac{1 - \beta^{n_0}}{n_0} \quad \text{or, equivalently,} \quad \frac{1 - \beta^n}{1 - \beta^{n_0}} \leq \frac{n}{n_0}. \tag{87}$$

Hence

$$\frac{1}{\Gamma(p)} \int_{C_2}^\infty \mathbb{E}[e^{-\lambda V_n^*}] \lambda^{p-1} d\lambda \leq \left(\frac{n}{n_0} \right)^p \exp(-(n - n_0)(q + p \ln \beta)). \tag{88}$$

We conclude as above and use that $\frac{n}{n_0} \leq n - n_0 + 1$ which implies that

$$\frac{1}{\Gamma(p)} \int_{C_2}^\infty \mathbb{E}[e^{-\lambda V_n^*}] \lambda^{p-1} d\lambda \leq (n - n_0 + 1)^p e^{-\rho(n-n_0)} \tag{89}$$

Thus together with (84) we get that

$$\frac{1}{\Gamma(p)} \int_{C_1}^\infty \mathbb{E}[e^{-\lambda V_n^*}] \lambda^{p-1} d\lambda \leq D_{\rho,p}. \tag{90}$$

3) Finally, let $\lambda \in [0, C_1]$. Then none of the summands are capped, and therefore

$$\begin{aligned} -\ln \mathbb{E} \left[e^{-\lambda V_n^*} \right] &\geq \sum_{k=n_0+1}^n \frac{1-\beta}{1-\beta^n} \beta^{n-k} \lambda + \frac{1-\beta^{n_0}}{1-\beta^n} \beta^{n-n_0} \lambda \\ &= \lambda \frac{1-\beta^{n-n_0}}{1-\beta^n} + \lambda \frac{\beta^{n-n_0}-\beta^n}{1-\beta^n} = \lambda. \end{aligned} \quad (91)$$

Thus

$$\frac{1}{\Gamma(p)} \int_0^{C_1} \mathbb{E}[e^{-\lambda V_n^*}] \lambda^{p-1} d\lambda \leq \frac{1}{\Gamma(p)} \int_0^\infty e^{-\lambda} \lambda^{p-1} d\lambda = 1. \quad (92)$$

Note that the first summand of the series defining $D_{\rho,p}$ is one and thus we get with (84) and (89) that

$$\frac{1}{\Gamma(p)} \int_0^\infty \mathbb{E}[e^{-\lambda V_n^*}] \lambda^{p-1} d\lambda \leq \sum_{r=0}^{n-n_0} (r+1)^p e^{-\rho r} \leq D_{\rho,p}. \quad (93)$$

□

Lemma 4.5. *Let $c, q, \rho \in (0, \infty)$, $p \in [1, \infty)$, $\beta \in (0, 1)$ satisfy*

$$q + p \ln \beta \geq \rho, \quad (94)$$

let $\mathcal{V} \subseteq \mathbb{R}^d$ be measurable, and let (X, U) be (c, q) -regular on $\mathcal{V}$ in the sense of Definition 4.1. Consider for each $i \in \{1, \dots, d\}$ the stopping time

$$\bar{\tau} = \inf \{n \in \mathbb{N}_0 : \theta_n \notin \mathcal{V}\} \quad \text{and} \quad \sigma^{(i)} = \inf \{n \in \mathbb{N} : v_n^{(i)} \geq c\}. \quad (95)$$

Then for every $i \in \{1, \dots, d\}$, $n \in \mathbb{N}$ one has

$$\mathbb{E} \left[(v_n^{(i)})^{-p} \mathbb{1}_{\{\sigma^{(i)} \leq n\}} \mathbb{1}_{\{\bar{\tau} \geq n\}} \right] \leq c^{-p} D_{\rho,p}, \quad (96)$$

where $D_{\rho,p}$ is given by (69).

Proof. Fix $i \in \{1, \dots, d\}$ and consider for all $n \in \mathbb{N}$

$$V_n^{(i)} = v_n^{(i)} + \mathbb{1}_{\{\bar{\tau} < n\}} \infty. \quad (97)$$

We use the recursion (4) to show that by induction for $m \in \{1, \dots, n\}$ one has

$$-\ln \mathbb{E}[\exp\{-\lambda V_n^{(i)}\} | \mathcal{F}_m] \geq \sum_{k=m+1}^n \left(c \frac{1-\beta}{1-\beta^n} \beta^{n-k} \lambda \right) \wedge q + \frac{1-\beta^m}{1-\beta^n} \beta^{n-m} \lambda V_m^{(i)}. \quad (98)$$

Indeed, if we set

$$a_n = \frac{\beta(1-\beta^{n-1})}{1-\beta^n} \quad \text{and} \quad b_n = \frac{1-\beta}{1-\beta^n}, \quad (99)$$

then on the event $\{\bar{\tau} \geq n\}$ one has

$$V_n^{(i)} = a_n V_{n-1}^{(i)} + b_n X^{(i)}(\theta_{n-1}, U_n)^2. \quad (100)$$

Since both sides vanish on the complementary event after applying $\exp(-\lambda \cdot)$, this implies

$$\begin{aligned} \mathbb{E}[e^{-\lambda V_n^{(i)}} | \mathcal{F}_{n-1}] &= e^{-a_n \lambda V_{n-1}^{(i)}} \mathbb{1}_{\{\bar{\tau} \geq n\}} \mathbb{E} \left[e^{-b_n \lambda X^{(i)}(\theta_{n-1}, U_n)^2} \mid \mathcal{F}_{n-1} \right] \\ &\leq e^{-a_n \lambda V_{n-1}^{(i)}} \exp(-(cb_n \lambda) \wedge q), \end{aligned} \quad (101)$$

where we used the (c, q) -regularity on the event $\{\bar{\tau} \geq n\} \subseteq \{\theta_{n-1} \in \mathcal{V}\}$. Therefore,

$$-\ln \mathbb{E}[e^{-\lambda V_n^{(i)}} | \mathcal{F}_{n-1}] \geq \left(c \frac{1-\beta}{1-\beta^n} \lambda \right) \wedge q + \frac{\beta(1-\beta^{n-1})}{1-\beta^n} \lambda V_{n-1}^{(i)}, \quad (102)$$

and iteration yields the displayed estimate, since

$$\prod_{j=m+1}^n \frac{\beta(1-\beta^{j-1})}{1-\beta^j} = \frac{1-\beta^m}{1-\beta^n} \beta^{n-m}. \quad (103)$$

Consequently, on the event $\{\sigma^{(i)} = m\}$, the bound $V_m^{(i)} \geq c$ holds, with the value $+\infty$ allowed if $\bar{\tau} < m$ and, hence, on this event

$$-\ln \mathbb{E}[\exp\{-\lambda V_n^{(i)}\} | \mathcal{F}_m] \geq \sum_{k=m+1}^n \left(c \frac{1-\beta}{1-\beta^n} \beta^{n-k} \lambda \right) \wedge q + c \frac{1-\beta^m}{1-\beta^n} \beta^{n-m} \lambda. \quad (104)$$

If $m = n$, then on $\{\sigma^{(i)} = m\}$ we have $v_n^{(i)} \geq c$. Since the first summand in the definition of $D_{\rho,p}$ is equal to one, this gives on the latter event that

$$\mathbb{E}[\mathbb{1}_{\{\bar{\tau} \geq n\}} (v_n^{(i)})^{-p} | \mathcal{F}_n] = \mathbb{1}_{\{\bar{\tau} \geq n\}} (v_n^{(i)})^{-p} \leq c^{-p} \leq c^{-p} D_{\rho,p}. \quad (105)$$

If $m < n$, then we apply [Lemma 4.4](#) for the conditional distribution of $V_n^{(i)}$ given $\mathcal{F}_m$ and get that on $\{\sigma^{(i)} = m\}$

$$\mathbb{E}[\mathbb{1}_{\{\bar{\tau} \geq n\}} (v_n^{(i)})^{-p} | \mathcal{F}_m] = \mathbb{E}[(V_n^{(i)})^{-p} | \mathcal{F}_m] \leq c^{-p} D_{\rho,p}. \quad (106)$$

The statement follows by decomposing the event $\{\sigma^{(i)} \leq n\}$ according to the hitting time:

$$\mathbb{E}[\mathbb{1}_{\{\sigma^{(i)} \leq n\}} \mathbb{1}_{\{\bar{\tau} \geq n\}} (v_n^{(i)})^{-p}] = \sum_{m=1}^n \mathbb{E}[\mathbb{1}_{\{\sigma^{(i)} = m\}} \mathbb{E}[\mathbb{1}_{\{\bar{\tau} \geq n\}} (v_n^{(i)})^{-p} | \mathcal{F}_m]] \leq c^{-p} D_{\rho,p}. \quad (107)$$

□

5. TECHNICAL ESTIMATES FOR THE PROPERTY IN [ITEM \(ii\)](#) OF [PROPOSITION 2.1](#)

This section controls the quadratic Taylor remainder appearing in [item \(ii\)](#) of [Proposition 2.1](#). The inverse squared conditioner is a decreasing function of the current squared innovation, which yields a nonpositive conditional covariance and permits the two factors to be separated. Combining this observation with a conditional second-moment bound for the innovation and a negative first-moment bound for v_{n-1} gives the required coordinatewise estimate.

Lemma 5.1. *One has for all $n \in \mathbb{N}$, $i \in \{1, \dots, d\}$ that*

$$\text{cov}([v_n^{(i)}]^{1/2} + \varepsilon)^{-2}, X^{(i)}(\theta_{n-1}, U_n)^2 | \mathcal{F}_{n-1} \leq 0 \quad (108)$$

on the event where the covariance is well-defined.

Proof. Fix $n \in \mathbb{N}$ and $i \in \{1, \dots, d\}$ and recall that

$$v_n^{(i)} = \frac{\beta(1-\beta^{n-1})}{1-\beta^n} v_{n-1}^{(i)} + \frac{1-\beta}{1-\beta^n} X^{(i)}(\theta_{n-1}, U_n)^2. \quad (109)$$

In terms of the non-negative random variables

$$A = \frac{\beta(1-\beta^{n-1})}{1-\beta^n} v_{n-1}^{(i)}, \quad B = \frac{1-\beta}{1-\beta^n} \quad \text{and} \quad Y = X^{(i)}(\theta_{n-1}, U_n)^2 \quad (110)$$

define the random $\mathcal{F}_{n-1}$ -measurable extended-valued function $G(x) = (\sqrt{A+Bx} + \varepsilon)^{-2}$ for $x \in [0, \infty)$, where the value is $+\infty$ if the denominator vanishes. Then

$$(\sqrt{v_n^{(i)}} + \varepsilon)^{-2} = \frac{1}{(\sqrt{A+BY} + \varepsilon)^2} = G(Y). \quad (111)$$

For every $m \in \mathbb{N}$, set

$$G_m(x) = \frac{1}{(\sqrt{A+Bx} + \varepsilon + m^{-1})^2}, \quad x \in [0, \infty). \quad (112)$$

The function G_m is bounded and decreasing. If Y' is conditionally on $\mathcal{F}_{n-1}$ independent of Y and identically distributed as Y , then

$$2\text{cov}(G_m(Y), Y|\mathcal{F}_{n-1}) = \mathbb{E}[(G_m(Y) - G_m(Y'))(Y - Y')|\mathcal{F}_{n-1}] \leq 0. \quad (113)$$

On the event where $\text{cov}(G(Y), Y|\mathcal{F}_{n-1})$ is well-defined, the random variables entering its definition are integrable. Since $G_m \uparrow G$, conditional monotone convergence yields

$$\text{cov}(G(Y), Y|\mathcal{F}_{n-1}) = \lim_{m \rightarrow \infty} \text{cov}(G_m(Y), Y|\mathcal{F}_{n-1}) \leq 0. \quad (114)$$

This finishes the proof. $\square$

Proposition 5.2. *Let $\beta \in [1/2, 1)$, $\varepsilon \in [0, \infty)$, $n \in \mathbb{N} \setminus \{1\}$, $i \in \{1, \dots, d\}$, let $A_n \in \mathcal{F}_{n-1}$, and let $C_X^{(2)}, C_v^{(-1)} \in (0, \infty)$. Assume that*

$$\mathbb{E}[(X^{(i)}(\theta_{n-1}, U_n))^2|\mathcal{F}_{n-1}] \leq C_X^{(2)} \quad \text{on } A_n \quad (115)$$

and

$$\mathbb{E}[\mathbb{1}_{A_n}(v_{n-1}^{(i)})^{-1}] \leq C_v^{(-1)}. \quad (116)$$

The negative-moment assumption implies that $v_{n-1}^{(i)} > 0$ almost surely on A_n . If $\varepsilon = 0$, inverse factors used in the proof are evaluated on A_n and extended by zero to A_n^c . Then one has

$$\mathbb{E}[\mathbb{1}_{A_n}(\sqrt{v_n^{(i)}} + \varepsilon)^{-2} X^{(i)}(\theta_{n-1}, U_n)^2] \leq 4C_X^{(2)} C_v^{(-1)}. \quad (117)$$

Proof. The negative-moment assumption implies that $v_{n-1}^{(i)} > 0$ almost surely on A_n . Since $\beta \geq \frac{1}{2}$ and $n \geq 2$, the recursion for v_n also gives $v_n^{(i)} > 0$ almost surely on A_n . We extend all inverse factors by zero to A_n^c . By Lemma 5.1, one has

$$\mathbb{1}_{A_n} \mathbb{E}[(\sqrt{v_n^{(i)}} + \varepsilon)^{-2} X^{(i)}(\theta_{n-1}, U_n)^2|\mathcal{F}_{n-1}] \leq \mathbb{1}_{A_n} \mathbb{E}[(\sqrt{v_n^{(i)}} + \varepsilon)^{-2}|\mathcal{F}_{n-1}] \mathbb{E}[X^{(i)}(\theta_{n-1}, U_n)^2|\mathcal{F}_{n-1}]. \quad (118)$$

The conditional second-moment assumption implies that, on A_n ,

$$\mathbb{E}[X^{(i)}(\theta_{n-1}, U_n)^2|\mathcal{F}_{n-1}] \leq C_X^{(2)}. \quad (119)$$

Since $\frac{\beta(1-\beta^{n-1})}{1-\beta^n} \leq 1$, the recursion for v_n yields

$$v_n^{(i)} \geq \frac{\beta(1-\beta^{n-1})}{1-\beta^n} v_{n-1}^{(i)}. \quad (120)$$

Thus

$$\begin{aligned} \mathbb{1}_{A_n} \mathbb{E}[(\sqrt{v_n^{(i)}} + \varepsilon)^{-2}|\mathcal{F}_{n-1}] &\leq \mathbb{1}_{A_n} \mathbb{E}[(v_n^{(i)})^{-1}|\mathcal{F}_{n-1}] \\ &\leq \mathbb{1}_{A_n} \frac{1-\beta^n}{\beta(1-\beta^{n-1})} (v_{n-1}^{(i)})^{-1}. \end{aligned} \quad (121)$$

Lemma 4.3 and $\beta \geq \frac{1}{2}$ imply that

$$\frac{1-\beta^n}{\beta(1-\beta^{n-1})} \leq \frac{1}{\beta} \frac{n}{n-1} \leq 4. \quad (122)$$

Combining (118), (119), (121), and (122) gives

$$\mathbb{E}[\mathbb{1}_{A_n}(\sqrt{v_n^{(i)}} + \varepsilon)^{-2} X^{(i)}(\theta_{n-1}, U_n)^2] \leq 4C_X^{(2)} \mathbb{E}[\mathbb{1}_{A_n}(v_{n-1}^{(i)})^{-1}] \leq 4C_X^{(2)} C_v^{(-1)}. \quad (123)$$

This proves (117). $\square$

6. INITIALISATION ESTIMATES

This section estimates the time required for each coordinate of the conditioner to enter the regime where negative moments are available. We first convert the conditional Laplace-transform bounds into lower-tail estimates and then iterate these estimates over blocks whose lengths are comparable to the memory scale $(1-\beta)^{-1}$. This yields an exponential bound, uniform in β , for the probability that a coordinate has not started before the process exits the prescribed region.

Lemma 6.1. *Let $c, q \in (0, \infty)$, $\beta \in (0, 1)$, $n \in \mathbb{N}$. Let V_n^* be a nonnegative random variable which satisfies for all $\lambda \in [0, \infty)$ that*

$$-\ln \mathbb{E}[e^{-\lambda V_n^*}] \geq \sum_{k=1}^n \min\left\{q, c \frac{1-\beta}{1-\beta^n} \beta^{n-k} \lambda\right\}. \quad (124)$$

Then

$$\mathbb{P}(V_n^* \leq c/2) \leq \exp\left(-\frac{q}{2} \frac{1-\beta^n}{1-\beta}\right). \quad (125)$$

Proof. Set

$$\lambda_* = \frac{q(1-\beta^n)}{c(1-\beta)}. \quad (126)$$

By the exponential Chebyshev inequality, we have that

$$\mathbb{P}(V_n^* \leq c/2) \leq e^{\lambda_* c/2} \mathbb{E}[e^{-\lambda_* V_n^*}]. \quad (127)$$

By (124),

$$\mathbb{E}[e^{-\lambda_* V_n^*}] \leq \exp\left(-\sum_{k=1}^n \left(c \frac{1-\beta}{1-\beta^n} \beta^{n-k} \lambda_*\right) \wedge q\right). \quad (128)$$

Now, for every $k \in \{1, \dots, n\}$, one has

$$c \frac{1-\beta}{1-\beta^n} \beta^{n-k} \lambda_* = q \beta^{n-k} \leq q. \quad (129)$$

Hence all terms inside the minimum are unsaturated and therefore

$$\sum_{k=1}^n \left(c \frac{1-\beta}{1-\beta^n} \beta^{n-k} \lambda_*\right) \wedge q = q \sum_{k=1}^n \beta^{n-k} = q \frac{1-\beta^n}{1-\beta}. \quad (130)$$

Consequently, we get with (127) that

$$\mathbb{P}(V_n^* \leq c/2) \leq \exp\left(\lambda_* c/2 - q \frac{1-\beta^n}{1-\beta}\right) = \exp\left(-\frac{q}{2} \frac{1-\beta^n}{1-\beta}\right), \quad (131)$$

which proves the claim. $\square$

Lemma 6.2. *Let $c, q \in (0, \infty)$, $\beta \in (0, 1)$, $n_0, n \in \mathbb{N}$ with $n_0 < n$. Let V_n^* be a nonnegative random variable which satisfies for all $\lambda \in [0, \infty)$ that*

$$-\ln \mathbb{E}[e^{-\lambda V_n^*}] \geq \sum_{k=n_0+1}^n \min\left\{q, c \frac{1-\beta}{1-\beta^n} \beta^{n-k} \lambda\right\}. \quad (132)$$

Then

$$\mathbb{P}\left(V_n^* \leq \frac{c}{2} \frac{1-\beta^{n-n_0}}{1-\beta^n}\right) \leq \exp\left(-\frac{q}{2} \frac{1-\beta^{n-n_0}}{1-\beta}\right). \quad (133)$$

If, in addition, $n - n_0 \geq \frac{1}{2}(1-\beta)^{-1}$, then

$$\mathbb{P}\left(V_n^* \leq \frac{c}{2} (1 - e^{-1/2})\right) \leq \exp\left(-\frac{q}{2} \frac{1 - e^{-1/2}}{1-\beta}\right). \quad (134)$$

Proof. Set

$$\lambda_* = \frac{q(1 - \beta^n)}{c(1 - \beta)}. \quad (135)$$

By the exponential Chebyshev inequality, we obtain

$$\mathbb{P}\left(V_n^* \leq \frac{c}{2} \frac{1 - \beta^{n-n_0}}{1 - \beta^n}\right) \leq \exp\left(\lambda_* \frac{c}{2} \frac{1 - \beta^{n-n_0}}{1 - \beta^n}\right) \mathbb{E}[e^{-\lambda_* V_n^*}]. \quad (136)$$

By (132),

$$\mathbb{E}[e^{-\lambda_* V_n^*}] \leq \exp\left(-\sum_{k=n_0+1}^n \left(c \frac{1 - \beta}{1 - \beta^n} \beta^{n-k} \lambda_*\right) \wedge q\right). \quad (137)$$

Now, for every $k \in \{n_0 + 1, \dots, n\}$, one has

$$c \frac{1 - \beta}{1 - \beta^n} \beta^{n-k} \lambda_* = q \beta^{n-k} \leq q. \quad (138)$$

Hence all terms inside the minimum are unsaturated and therefore

$$\sum_{k=n_0+1}^n \left(c \frac{1 - \beta}{1 - \beta^n} \beta^{n-k} \lambda_*\right) \wedge q = q \sum_{k=n_0+1}^n \beta^{n-k} = q \frac{1 - \beta^{n-n_0}}{1 - \beta}. \quad (139)$$

Consequently,

$$\begin{aligned} \mathbb{P}\left(V_n^* \leq \frac{c}{2} \frac{1 - \beta^{n-n_0}}{1 - \beta^n}\right) &\leq \exp\left(\lambda_* \frac{c}{2} \frac{1 - \beta^{n-n_0}}{1 - \beta^n} - q \frac{1 - \beta^{n-n_0}}{1 - \beta}\right) \\ &= \exp\left(-\frac{q}{2} \frac{1 - \beta^{n-n_0}}{1 - \beta}\right), \end{aligned} \quad (140)$$

which proves the first assertion.

For the second assertion, note that $n - n_0 \geq \frac{1}{2}(1 - \beta)^{-1}$ and hence

$$\beta^{n-n_0} \leq e^{-(1-\beta)(n-n_0)} \leq e^{-1/2}. \quad (141)$$

Therefore

$$\frac{1 - \beta^{n-n_0}}{1 - \beta^n} \geq 1 - \beta^{n-n_0} \geq 1 - e^{-1/2} \quad (142)$$

and

$$\left\{V_n^* \leq \frac{c}{2}(1 - e^{-1/2})\right\} \subseteq \left\{V_n^* \leq \frac{c}{2} \frac{1 - \beta^{n-n_0}}{1 - \beta^n}\right\}. \quad (143)$$

Applying the first estimate finishes the proof:

$$\mathbb{P}\left(V_n^* \leq \frac{c}{2}(1 - e^{-1/2})\right) \leq \exp\left(-\frac{q}{2} \frac{1 - e^{-1/2}}{1 - \beta}\right). \quad (144)$$

□

Proposition 6.3. *Let $c, q \in (0, \infty)$, $\beta \in [1/2, 1)$, let $\mathcal{V} \subseteq \mathbb{R}^d$ be measurable, and let (X, U) be (c, q) -regular on $\mathcal{V}$ in the sense of Definition 4.1. Consider the stopping time*

$$\bar{\tau} = \inf\{n \in \mathbb{N}_0 : \theta_n \notin \mathcal{V}\} \quad \text{and} \quad \sigma^{(i)} = \inf\{n \in \mathbb{N} : v_n^{(i)} \geq \frac{c(1 - e^{-1/2})}{2}\}. \quad (145)$$

Then for all $n \in \mathbb{N}$ one has

$$\mathbb{P}(\bar{\tau} \geq n, \sigma^{(i)} > n) \leq \exp\left(-\frac{qn(1 - e^{-1/2})}{4}\right). \quad (146)$$

Proof. Set

$$\eta = \exp\left(-\frac{q}{2} \frac{1 - e^{-1/2}}{1 - \beta}\right). \quad (147)$$

First suppose that $n \leq (1 - \beta)^{-1}$. Then [Proposition 4.2](#) with $n_0 = 0$ and [Lemma 6.1](#) imply that

$$\begin{aligned} \mathbb{P}(\bar{\tau} \geq n, \sigma^{(i)} > n) &\leq \mathbb{P}\left(\bar{\tau} \geq n, v_n^{(i)} \leq \frac{1 - e^{-1/2}}{2} c\right) \\ &\leq \mathbb{P}\left(\bar{\tau} \geq n, v_n^{(i)} \leq \frac{c}{2}\right) \leq \exp\left(-\frac{q}{2} \frac{1 - \beta^n}{1 - \beta}\right). \end{aligned} \quad (148)$$

For $n = 1$, this already implies the claim. Thus we may assume that $n \geq 2$. Since

$$\frac{1 - \beta^n}{1 - \beta} = \sum_{k=0}^{n-1} \beta^k \geq n\beta^{n-1}, \quad (149)$$

and since $n \leq (1 - \beta)^{-1}$ implies $\beta \geq 1 - \frac{1}{n}$, we get for $n \geq 2$ that

$$\beta^{n-1} \geq \left(1 - \frac{1}{n}\right)^{n-1} = \left(\left(1 + \frac{1}{n-1}\right)^{n-1}\right)^{-1} \geq e^{-1} \geq \frac{1}{4}. \quad (150)$$

Consequently, by (148), (149), and (150),

$$\mathbb{P}(\bar{\tau} \geq n, \sigma^{(i)} > n) \leq \exp\left(-\frac{q}{8} n\right) \leq \exp\left(-\frac{q(1 - e^{-1/2})}{4} n\right). \quad (151)$$

Now suppose that $n > (1 - \beta)^{-1}$ and set

$$L = \left\lceil \frac{1}{2(1 - \beta)} \right\rceil \quad \text{and} \quad J = \left\lfloor \frac{n}{L} \right\rfloor. \quad (152)$$

Since $\beta \geq \frac{1}{2}$, one has

$$\frac{1}{2(1 - \beta)} \leq L \leq \frac{1}{1 - \beta}. \quad (153)$$

Moreover, $J \geq 1$ and

$$J \geq \frac{n}{2L} \geq \frac{n(1 - \beta)}{2}. \quad (154)$$

Setting, for all $j = 2, \dots, J$,

$$t_0 = 0 \quad \text{and} \quad t_j = n - JL + jL, \quad (155)$$

we obtain a partition

$$0 = t_0 < t_1 < \dots < t_J = n \quad (156)$$

such that all $j = 1, \dots, J$

$$t_j - t_{j-1} \geq L \geq \frac{1}{2(1 - \beta)}. \quad (157)$$

We now show by induction that

$$\mathbb{P}(\bar{\tau} \geq t_j, \sigma^{(i)} > t_j) \leq \eta^j, \quad j \in \{1, \dots, J\}. \quad (158)$$

For $j = 1$, [Proposition 4.2](#) with $n_0 = 0$ and [Lemma 6.1](#) give

$$\begin{aligned} \mathbb{P}(\bar{\tau} \geq t_1, \sigma^{(i)} > t_1) &\leq \mathbb{P}\left(\bar{\tau} \geq t_1, v_{t_1}^{(i)} \leq \frac{1 - e^{-1/2}}{2} c\right) \\ &\leq \mathbb{P}\left(\bar{\tau} \geq t_1, v_{t_1}^{(i)} \leq \frac{c}{2}\right) \leq \exp\left(-\frac{q}{2} \frac{1 - \beta^{t_1}}{1 - \beta}\right). \end{aligned} \quad (159)$$

Since $t_1 \geq \frac{1}{2(1-\beta)}$, we have

$$\beta^{t_1} = \exp\{t_1 \ln \beta\} \leq \exp\{-t_1(1-\beta)\} \leq e^{-1/2} \quad (160)$$

and hence

$$\mathbb{P}(\bar{\tau} \geq t_1, \sigma^{(i)} > t_1) \leq \eta. \quad (161)$$

Next let $j \in \{2, \dots, J\}$ and assume that

$$\mathbb{P}(\bar{\tau} \geq t_{j-1}, \sigma^{(i)} > t_{j-1}) \leq \eta^{j-1}. \quad (162)$$

Set $A_{j-1} = \{\bar{\tau} > t_{j-1}, \sigma^{(i)} > t_{j-1}\}$. Then

$$\mathbb{P}(\bar{\tau} \geq t_j, \sigma^{(i)} > t_j) = \mathbb{E}\left[\mathbb{1}_{A_{j-1}} \mathbb{P}(\bar{\tau} \geq t_j, \sigma^{(i)} > t_j \mid \mathcal{F}_{t_{j-1}})\right]. \quad (163)$$

Define the shifted exit time

$$\bar{\tau}^{(j)} = \inf\{m \in \mathbb{N}_0 : m \geq t_{j-1} \text{ and } \theta_m \notin \mathcal{V}\}. \quad (164)$$

On A_{j-1} this shifted stopping time agrees with $\bar{\tau}$ up to the interval $[t_{j-1}, t_j]$. [Proposition 4.2](#), applied conditionally on $\mathcal{F}_{t_{j-1}}$ with initial time t_{j-1} and exit time $\bar{\tau}^{(j)}$, implies that the random variable

$$V_{t_j}^* = v_{t_j}^{(i)} + \mathbb{1}_{\{t_j \leq \bar{\tau}^{(j)}\}} c \infty \quad (165)$$

satisfies the Laplace bound in [Lemma 6.2](#). Since

$$t_j - t_{j-1} \geq \frac{1}{2(1-\beta)}, \quad (166)$$

[Lemma 6.2](#) yields, on A_{j-1} ,

$$\mathbb{P}\left(V_{t_j}^* \leq \frac{c}{2}(1 - e^{-1/2}) \mid \mathcal{F}_{t_{j-1}}\right) \leq \eta. \quad (167)$$

Moreover, on A_{j-1} ,

$$\{\bar{\tau} \geq t_j, \sigma^{(i)} > t_j\} \subseteq \left\{V_{t_j}^* \leq \frac{c}{2}(1 - e^{-1/2})\right\}. \quad (168)$$

Consequently, by (163), (167), and (168),

$$\mathbb{P}(\bar{\tau} \geq t_j, \sigma^{(i)} > t_j) \leq \eta \mathbb{P}(A_{j-1}) \leq \eta \mathbb{P}(\bar{\tau} \geq t_{j-1}, \sigma^{(i)} > t_{j-1}) \leq \eta^j. \quad (169)$$

This proves (158).

Since $t_J = n$, we conclude with (154) that

$$\mathbb{P}(\bar{\tau} \geq n, \sigma^{(i)} > n) \leq \eta^J \leq \exp\left(-\frac{q}{2} \frac{1 - e^{-1/2}}{1 - \beta} \frac{n(1 - \beta)}{2}\right) = \exp\left(-\frac{q(1 - e^{-1/2})}{4} n\right), \quad (170)$$

which proves the claim. $\square$

7. COMBINING THE ESTIMATES BEFORE AND AFTER THE COORDINATE STARTUP TIMES

This section assembles the preceding estimates into the final error bound. At every iteration, each coordinate is split according to whether its startup time has occurred: the negative-moment, covariance, and quadratic estimates are used in the late regime, while direct self-normalization and the startup tail bound control the early regime. Together, these estimates yield the general recursion and, after convolution in numerical time, the main theorem with constants uniform in β and ε .

Theorem 7.1. *Let $\beta \in [1/2, 1)$, $\varepsilon \in [0, \infty)$, $\rho, c, q, C_X, L_f, C_{\text{Loj}} \in (0, \infty)$, let $V \subseteq \mathbb{R}^d$ be measurable, and suppose that the following is true:*

(a) *It holds that*

$$q + 3 \ln \beta \geq \rho. \quad (171)$$

(b) (X, U) is (c, q) -regular on V in the sense of [Definition 4.1](#) and uniformly bounded: for every $\theta \in V$, $i \in \{1, \dots, d\}$ one has

$$\|X^{(i)}(\theta, U)\|_{L^\infty} \leq C_X. \quad (172)$$

(c) There exists $F: \mathbb{R}^d \rightarrow [0, \infty)$ with globally L_f -Lipschitz continuous derivative such that for every $\theta \in V$ one has

$$f_X(\theta) = -\nabla F(\theta) \quad (173)$$

and

$$F(\theta) \leq C_{\text{Loj}} |f_X(\theta)|^2. \quad (174)$$

(d) The step-size sequence $(\gamma_n)_{n \in \mathbb{N}}$ is $(0, \infty)$ -valued and satisfies for all $n \in \mathbb{N} \setminus \{1\}$ that

$$\gamma_n \leq C_{\text{Loj}}(C_X + \varepsilon). \quad (175)$$

Set

$$\begin{aligned} \kappa_0 &= \frac{4}{1 - e^{-1/2}}, & \kappa_1 &= \frac{1}{2C_{\text{Loj}}(C_X + \varepsilon)}, \\ \kappa_2 &= \frac{8\kappa_0^3 d L_f C_X^6 C_{\text{Loj}}(C_X + \varepsilon)}{c^3} D_{\rho, 3}, & \kappa_3 &= \frac{\kappa_0 d L_f C_X^2}{c} D_{\rho, 1}, \end{aligned} \quad (176)$$

and, for every $n \in \mathbb{N} \setminus \{1\}$,

$$R_n^{\text{early}} = d \left(2C_X n^{1/2} + \frac{L_f}{2} n \gamma_n \right) e^{-q(n-1)/\kappa_0} \gamma_n. \quad (177)$$

For every $n \in \mathbb{N}_0$ set $t_n = \sum_{k=1}^n \gamma_k$. Then one has for every $n \in \mathbb{N}$ that

$$\begin{aligned} &\mathbb{E}[F(\theta_n) \mathbb{1}_{\{\tau \geq n\}}] \\ &\leq [F(\theta_0)^{1/2} + (2^{-1} d L_f)^{1/2} \gamma_1]^2 e^{\kappa_1(t_1 - t_n)} + \sum_{j=2}^n \left[\kappa_2 [(j-1)^{-1} \vee (1-\beta)]^2 \gamma_j + \kappa_3 \gamma_j^2 + R_j^{\text{early}} \right] e^{\kappa_1(t_j - t_n)}, \end{aligned} \quad (178)$$

where $\tau = \inf\{n \in \mathbb{N}_0 : \theta_n \notin V\}$.

Proof. If $\theta_0 \notin V$, then $\tau = 0$ and the left-hand side of (178) vanishes. Hence we may assume without loss of generality that $\theta_0 \in V$. Set

$$c_v = (C_X + \varepsilon)^{-1}, \quad c_{\text{Loj}} = (2L_f)^{-1}. \quad (179)$$

For every $n \in \mathbb{N}_0$ set

$$\varphi_n = \mathbb{E}[F(\theta_n) \mathbb{1}_{\{\tau \geq n\}}]. \quad (180)$$

For every $x \in \mathbb{R}^d$, the L_f -smoothness inequality, applied with $h = -L_f^{-1} \nabla F(x)$, and the nonnegativity of F give

$$0 \leq F(x - L_f^{-1} \nabla F(x)) \leq F(x) - \frac{1}{2L_f} |\nabla F(x)|^2, \quad |\nabla F(x)|^2 \leq 2L_f F(x). \quad (181)$$

Thus the choice of c_{Loj} in (179) is admissible by (173) and (181). For every $i \in \{1, \dots, d\}$ define $\sigma^{(i)} = \inf\{n \in \mathbb{N} : v_n^{(i)} \geq 2c/\kappa_0\}$. For every $n \geq 2$ set

$$\begin{aligned} h_n &= (n-1)^{-1} \vee (1-\beta), & p_n &= e^{-q(n-1)/\kappa_0}, & A_n &= \{\tau \geq n\}, \\ G_{n,i} &= \{\sigma^{(i)} \leq n-1\}, & Y_{n,i} &= (\sqrt{v_n^{(i)}} + \varepsilon)^{-1} X^{(i)}(\theta_{n-1}, U_n). \end{aligned} \quad (182)$$

By (182), the events A_n and $G_{n,i}$ are $\mathcal{F}_{n-1}$ -measurable. On $G_{n,i}$, let

$$\xi_{n,i} = \text{cov}((\sqrt{v_n^{(i)}} + \varepsilon)^{-1}, X^{(i)}(\theta_{n-1}, U_n) \mid \mathcal{F}_{n-1}). \quad (183)$$

When $\varepsilon = 0$, inverse factors and covariances used with $\mathbb{1}_{G_{n,i}}$ are extended by zero to $G_{n,i}^c$.

Step 1. Late coordinates satisfy uniform moment bounds, while early coordinates are exponentially rare.

On A_n , (4), (172), and (182) give $0 \leq v_n^{(i)} \leq C_X^2$; hence, by (179), $(\sqrt{v_n^{(i)}} + \varepsilon)^{-1} \geq c_v$ on $A_n \cap G_{n,i}$. Put $\tilde{c} = 2c/\kappa_0$. Since $\tilde{c} < c$, (X, U) is also $(\tilde{c}, q)$ -regular, and the corresponding startup time in Lemma 4.5 is $\sigma^{(i)}$. Since $A_n \cap G_{n,i} \subseteq \{\tau \geq n-1, \sigma^{(i)} \leq n-1\}$ and since $q + r \ln \beta \geq \rho$ for every $r \in \{1, 3\}$, Lemma 4.5 gives, for every $r \in \{1, 3\}$,

$$\mathbb{E}[\mathbb{1}_{A_n \cap G_{n,i}} (v_{n-1}^{(i)})^{-r}] \leq \tilde{c}^r D_{\rho, r}. \quad (184)$$

Moreover,

$$\frac{1 - \beta}{1 - \beta^{n-1}} \leq 2h_n \quad (185)$$

follows from Lemma 3.2 and (182). Combining (184) and (185) with Proposition 5.2 and Proposition 3.1 and (172) yields

$$\mathbb{E}\left[\mathbb{1}_{A_n} \sum_{i=1}^d \mathbb{1}_{G_{n,i}} Y_{n,i}^2\right] \leq \frac{2d\kappa_0 C_X^2}{c} D_{\rho, 1}, \quad \mathbb{E}\left[\mathbb{1}_{A_n} \sum_{i=1}^d \mathbb{1}_{G_{n,i}} \xi_{n,i}^2\right]^{1/2} \leq 8\sqrt{d} C_X^3 \left(\frac{\kappa_0}{2c}\right)^{3/2} D_{\rho, 3}^{1/2} h_n. \quad (186)$$

Moreover, (4) and (182) give $|Y_{n,i}| \leq n^{1/2}$, while Proposition 6.3 and (182) give $\mathbb{P}(A_n \cap G_{n,i}^c) \leq p_n$. Since $|f_X^{(i)}| \leq C_X$ on V , we obtain

$$\begin{aligned} \mathbb{E}\left[\mathbb{1}_{A_n} \sum_{i=1}^d \mathbb{1}_{G_{n,i}^c} |f_X^{(i)}(\theta_{n-1})|^2\right] &\leq dC_X^2 p_n, \quad \mathbb{E}\left[\mathbb{1}_{A_n} \sum_{i=1}^d \mathbb{1}_{G_{n,i}^c} |f_X^{(i)}(\theta_{n-1}) Y_{n,i}|\right] \leq dC_X n^{1/2} p_n, \\ \text{and} \quad \mathbb{E}\left[\mathbb{1}_{A_n} \sum_{i=1}^d \mathbb{1}_{G_{n,i}^c} Y_{n,i}^2\right] &\leq dnp_n. \end{aligned} \quad (187)$$

Step 2. The coordinate estimates and Proposition 2.1 yield a single Lyapunov recursion.

The lower bound for the inverse factor on $A_n \cap G_{n,i}$ proves condition (i) of Proposition 2.1. Conditions (ii) and (iii) are given by (186), while the required Łojasiewicz bounds follow from (174), (173), (181), and (179). Since $\theta_0 \in V$, the stopping times agree for $n_0 = 1$. Hence (22), applied with $n_0 = 1$, together with (187) and the definitions in (176), gives

$$\begin{aligned} \varphi_n &\leq \left(1 - \frac{c_v}{2C_{\text{Loj}}}\gamma_n\right) \mathbb{E}[\mathbb{1}_{A_n} F(\theta_{n-1})] + \kappa_2 h_n^2 \gamma_n + \kappa_3 \gamma_n^2 \\ &\quad + d(c_v C_X^2 + C_X n^{1/2}) p_n \gamma_n + \frac{dL_f}{2} n p_n \gamma_n^2. \end{aligned} \quad (188)$$

By (175), the coefficient in the first line is nonnegative, and $\mathbb{E}[\mathbb{1}_{A_n} F(\theta_{n-1})] \leq \varphi_{n-1}$. Since (179) gives $c_v C_X^2 \leq C_X \leq C_X n^{1/2}$, (177) now yields

$$\varphi_n \leq (1 - \kappa_1 \gamma_n) \varphi_{n-1} + \kappa_2 h_n^2 \gamma_n + \kappa_3 \gamma_n^2 + R_n^{\text{early}}. \quad (189)$$

Step 3. The scalar recursion can be iterated. For every $i \in \{1, \dots, d\}$, equations (4) and (5), applied at $n = 1$, give the first line below; together with (24), this yields the second:

$$\left|(\sqrt{v_1^{(i)}} + \varepsilon)^{-1} X^{(i)}(\theta_0, U_1)\right| \leq 1, \quad |\theta_1 - \theta_0| \leq \sqrt{d} \gamma_1. \quad (190)$$

Equations (23), (173), (181), and (190) yield

$$F(\theta_1) \leq F(\theta_0) + |\nabla F(\theta_0)| |\theta_1 - \theta_0| + \frac{L_f}{2} |\theta_1 - \theta_0|^2 \leq \left[F(\theta_0)^{1/2} + \left(\frac{dL_f}{2}\right)^{1/2} \gamma_1\right]^2. \quad (191)$$

By (180), $\varphi_1 \leq \mathbb{E}[F(\theta_1)]$; hence (191) controls φ_1 by the initial term in (178). Finally, (175) gives $\kappa_1 \gamma_n \leq 1/2$. Iterating (189) from time 1 and using $1 - x \leq e^{-x}$ proves (178). $\square$

Proof of Theorem 1.1. Fix $\varepsilon \in [0, 1]$. The constant κ_1 from Theorem 7.1 is

$$\kappa_1 = [2C_{\text{Loj}}(C_X + \varepsilon)]^{-1}. \quad (192)$$

The assumptions of Theorem 7.1 are satisfied. Since $\varepsilon \leq 1$, (7) and (192) give

$$\eta \leq \kappa_1 \leq [2C_{\text{Loj}}C_X]^{-1}. \quad (193)$$

Set $\bar{\gamma} = C_{\text{Loj}}(C_X + 1) = (2\eta)^{-1}$. Then the assumption that $(\gamma_n)_{n \in \mathbb{N}}$ is $(0, C_{\text{Lo}}(C_X + \varepsilon)]$ -valued and the fact that $\varepsilon \leq 1$ show that, for every $j \geq 2$,

$$\gamma_j \leq \bar{\gamma}. \quad (194)$$

By (176), the constant κ_2 satisfies

$$\kappa_2 \leq \frac{8\kappa_0^3 dL_f C_X^6 C_{\text{Loj}}(C_X + 1)}{c^3} D_{\rho,3} =: \bar{\kappa}_2, \quad (195)$$

whereas, by (176) and (177), κ_3 and R_j^{early} do not depend on ε . Since $\kappa_1 \geq \eta$ and $e^{\kappa_1 \gamma_1} e^{-\kappa_1 t_n} = e^{-\kappa_1(t_n - t_1)}$, the general estimate (178) implies, for every $n \geq 2$,

$$\begin{aligned} \mathbb{E}[\mathbb{1}_{\{\tau \geq n\}} F(\theta_n)] &\leq \left[F(\theta_0)^{1/2} + \left(\frac{dL_f}{2} \right)^{1/2} \gamma_1 \right]^2 e^{-\eta(t_n - t_1)} \\ &\quad + \sum_{j=2}^n \left[\kappa_2 \left(\frac{1}{j-1} \vee (1-\beta) \right)^2 \gamma_j + \kappa_3 \gamma_j^2 + R_j^{\text{early}} \right] e^{-\eta(t_n - t_j)}. \end{aligned} \quad (196)$$

For $a \in (0, \infty)$, the monotonicity of $(\gamma_j)_{j \in \mathbb{N}}$ and (194) give

$$\sum_{j=2}^n \gamma_j e^{-a(t_n - t_j)} \leq e^{a\bar{\gamma}} \sum_{j=2}^n \int_{t_{j-1}}^{t_j} e^{-a(t_n - s)} ds \leq \frac{e^{a\bar{\gamma}}}{a}. \quad (197)$$

Since $(\frac{1}{j-1} \vee (1-\beta))^2 \leq (j-1)^{-2} + (1-\beta)^2$, (11) and (197), applied with $a = \eta$, yield

$$\begin{aligned} \sum_{j=2}^n \left(\frac{1}{j-1} \vee (1-\beta) \right)^2 \gamma_j e^{-\eta(t_n - t_j)} &\leq (1-\beta)^2 \sum_{j=2}^n \gamma_j e^{-\eta(t_n - t_j)} + C_{\text{dec}} \gamma_n \sum_{j=2}^n \frac{e^{-\eta(t_n - t_j)/2}}{(j-1)^2} \\ &\leq \frac{e^{1/2}}{\eta} (1-\beta)^2 + C_{\text{dec}} \gamma_n \sum_{j=2}^n \frac{1}{(j-1)^2} \\ &= \frac{e^{1/2}}{\eta} (1-\beta)^2 + \frac{\pi^2}{6} C_{\text{dec}} \gamma_n. \end{aligned} \quad (198)$$

Similarly, (11) and (197), now applied with $a = \eta/2$, give

$$\sum_{j=2}^n \gamma_j^2 e^{-\eta(t_n - t_j)} \leq C_{\text{dec}} \gamma_n \sum_{j=2}^n \gamma_j e^{-\eta(t_n - t_j)/2} \leq \frac{2C_{\text{dec}} e^{1/4}}{\eta} \gamma_n. \quad (199)$$

Equations (177) and (194) give

$$R_j^{\text{early}} \leq d \left(2C_X j^{1/2} + \frac{L_f}{4\eta} j \right) e^{-q(j-1)/\kappa_0} \gamma_j. \quad (200)$$

The constant

$$C_{\text{early}}^{\text{time}} = dC_{\text{dec}} \sum_{j=2}^{\infty} \left(2C_X j^{1/2} + \frac{L_f}{4\eta} j \right) e^{-q(j-1)/\kappa_0} \quad (201)$$

is finite. Consequently, (11) and (200) imply

$$\sum_{j=2}^n R_j^{\text{early}} e^{-\eta(t_n - t_j)} \leq dC_{\text{dec}} \gamma_n \sum_{j=2}^n \left(2C_X j^{1/2} + \frac{L_f}{4\eta} j \right) e^{-q(j-1)/\kappa_0} e^{-\eta(t_n - t_j)/2} \leq C_{\text{early}}^{\text{time}} \gamma_n. \quad (202)$$

For $n = 1$, (12) follows directly from (178). For $n \geq 2$, equations (196), (195), (198), (199), and (201)–(202) prove (12) with the choices

$$C_0 = \frac{\pi^2}{6} \bar{\kappa}_2 C_{\text{dec}} + \frac{2\kappa_3 C_{\text{dec}} e^{1/4}}{\eta} + C_{\text{early}}^{\text{time}}, \quad C_1 = \frac{\bar{\kappa}_2 e^{1/2}}{\eta}, \quad (203)$$

which do not depend on ε or β . $\square$

Remark 7.2 (Non-asymptotic error analysis with fully explicit error constants). In Theorem 1.1 all constants, except of C_0 and C_1 , are explicitly specified and, actually, even C_0 and C_1 in Theorem 1.1 can be explicitly specified (see (203) in the proof of Theorem 1.1) and explicitly estimated from above. This is precisely the subject of this remark. More formally, combining (7), (69), (176), (195), (201), and (203) shows that

$$\begin{aligned} C_0 &= 6^{-1} \pi^2 \bar{\kappa}_2 C_{\text{dec}} + \eta^{-1} 2\kappa_3 C_{\text{dec}} e^{1/4} + C_{\text{early}}^{\text{time}} \\ &= 6^{-1} \pi^2 C_{\text{dec}} \left[\frac{8\kappa_0^3 L_f C_X^6 C_{\text{Loj}}(C_X + 1)d}{c^3} \right] D_{\rho,3} + \eta^{-1} 2\kappa_3 C_{\text{dec}} \exp(1/4) \\ &\quad + C_{\text{dec}} d \left[\sum_{j=2}^{\infty} \left(2C_X j^{1/2} + \frac{L_f j}{4\eta} \right) \exp(-q(j-1)/\kappa_0) \right]. \end{aligned} \quad (204)$$

Therefore, we obtain

$$\begin{aligned} C_0 &= 6^{-1} \pi^2 C_{\text{dec}} \left[\frac{4\kappa_0^3 L_f C_X^6 d}{c^3 \eta} \right] \left[\sum_{r=0}^{\infty} \frac{(r+1)^3}{\exp(\rho r)} \right] + \left[\frac{2\kappa_0 L_f C_X^2 d}{\eta c} \right] \left[\sum_{r=0}^{\infty} \frac{(r+1)}{\exp(\rho r)} \right] C_{\text{dec}} \exp(1/4) \\ &\quad + C_{\text{dec}} d \left[\sum_{j=2}^{\infty} \left(2C_X j^{1/2} + \frac{L_f j}{4\eta} \right) \exp(-q(j-1)/\kappa_0) \right]. \end{aligned} \quad (205)$$

Hence, we obtain

$$\begin{aligned} C_0 &= C_{\text{dec}} d \left[\sum_{r=0}^{\infty} \frac{2^7 \pi^2 c^{-3} \eta^{-1} L_f C_X^6 (r+1)^3}{3(1 - \exp(-1/2))^3 \exp(\rho r)} + \sum_{r=0}^{\infty} \frac{8\eta^{-1} c^{-1} L_f C_X^2 (r+1)}{(1 - \exp(-1/2)) \exp(\rho r - 1/4)} \right. \\ &\quad \left. + \sum_{j=2}^{\infty} \left(2C_X j^{1/2} + \frac{L_f j}{4\eta} \right) \exp(-q(j-1)(1 - \exp(-1/2))/4) \right]. \end{aligned} \quad (206)$$

Using $2^7 \pi^2 3^{-1} (1 - \exp(-1/2))^{-3} \leq 2^7 (63) = 2^7 (2^6 - 1) = 2^{13} - 2^7$, $(1 - \exp(-1/2))^{-1} \leq 4 = 2^2$, $j^{1/2} \leq j$, $[\exp(-1/4)]^{-1} \leq \exp(1/2) \leq 2$, $(1 - \exp(-1/2))/4 \geq 2^{-4}$, and the change of index $r = j - 1$ in the last series of (206), we obtain

$$C_0 \leq C_{\text{dec}} d (L_f + C_X + 1)^7 (\eta^{-1} + 1) \left[\sum_{r=0}^{\infty} \left(\frac{((2^{13} - 2^7) c^{-3} + 2^6 c^{-1})(r+1)^3}{\exp(\rho r)} + \frac{2^2 (r+1)}{\exp(2^{-4} q r)} \right) \right]. \quad (207)$$

Combining this with the fact that $2^6 c^{-1} + 2^2 \leq (2^6 + 2^2)(c^{-3} + 1) \leq 2^7 (c^{-3} + 1)$ and the fact that for all $r \in \mathbb{N}_0$ it holds that $(r+1) \leq (r+1)^3$ proves that

$$C_0 \leq 2^{13} C_{\text{dec}} d (L_f + C_X + 1)^7 (\eta^{-1} + 1) \left[\sum_{r=0}^{\infty} \frac{(c^{-3} + 1)(r+1)^3}{\exp(\min\{\rho, 2^{-4}q\}r)} \right]. \quad (208)$$

In addition, combining (7), (69), (176), (195), and (203) shows that

$$\begin{aligned} C_1 &= \bar{\kappa}_2 \eta^{-1} e^{1/2} = (c^{-3} 8\kappa_0^3 d L_f C_X^6 C_{\text{Loj}}(C_X + 1) D_{\rho,3}) \eta^{-1} \exp(1/2) = (c^{-3} 4\kappa_0^3 d L_f C_X^6 D_{\rho,3}) \eta^{-2} \exp(1/2) \\ &= \left[\frac{4\kappa_0^3 \exp(1/2) L_f C_X^6 d}{c^3 \eta^2} \right] \left[\sum_{r=0}^{\infty} (r+1)^3 e^{-\rho r} \right] = \left[\frac{4\kappa_0^3 \exp(1/2) L_f C_X^6 d}{c^3 \eta^2} \right] \left[\sum_{r=0}^{\infty} (r+1)^3 e^{-\rho r} \right] \\ &= \left[\frac{4^4 \exp(1/2) L_f C_X^6 d}{c^3 \eta^2 (1 - \exp(-1/2))^3} \right] \left[\sum_{r=0}^{\infty} \frac{(r+1)^3}{\exp(\rho r)} \right] = \left[\frac{2^8 \exp(\rho + 1/2) L_f C_X^6 d}{c^3 \eta^2 (1 - \exp(-1/2))^3} \right] \left[\sum_{r=1}^{\infty} \frac{r^3}{\exp(\rho r)} \right]. \end{aligned} \quad (209)$$

Using $\exp(1/2)[1 - \exp(-1/2)]^{-3} \leq 2^{15}$ we hence obtain that

$$C_1 = \left[\frac{2^8 \exp(\rho + 1/2) L_f C_X^6 d}{c^3 \eta^2 (1 - \exp(-1/2))^3} \right] \left[\sum_{r=0}^{\infty} \frac{r^3}{\exp(\rho r)} \right] \leq \sum_{r=0}^{\infty} \frac{2^{13} \eta^{-2} c^{-3} L_f C_X^6 d r^3}{\exp(\rho r - \rho)} = \sum_{r=0}^{\infty} \frac{2^{13} d L_f C_X^6 \eta^{-2} c^{-3} (r+1)^3}{\exp(\rho r)}. \quad (210)$$

In (206) and (210) we provide exact and explicit representations of the error constants C_0 and C_1 in (12) in Theorem 1.1 in terms of the parameters $d, \rho, c, q, C_X, L_f, C_{\text{dec}}, \eta$ in Theorem 1.1 (see (7)) and in (208) and (210) we provide short upper bounds for the error constants C_0 and C_1 in (12) in Theorem 1.1 in terms of the parameters $d, \rho, c, q, C_X, L_f, C_{\text{dec}}, \eta$ in Theorem 1.1 (see (7)).

8. CRITERIA AND EXAMPLES FOR REGULAR MINIBATCH INNOVATIONS

We conclude with sufficient conditions under which the regularity assumption in Definition 4.1 can be verified for minibatch stochastic gradients. We first reduce regularity to a uniform small-ball estimate. We then establish such an estimate for strongly convex coordinate gradients and apply the resulting criterion to a risk-sensitive objective. Finally, we treat nonlinear regression by a separate argument based on conditional density bounds and negative moments.

8.1. A small-ball criterion for regularity. Let $M \in \mathbb{N}$, let $U_1, \dots, U_M$ be independent copies of an $\mathbb{R}^m$ -valued random variable U , and set $\mathbf{U} = (U_1, \dots, U_M)$. For a loss $\ell: \mathbb{R}^d \times \mathbb{R}^m \rightarrow \mathbb{R}$ that is differentiable in its first argument, the natural minibatch innovation is defined, for every $\theta \in \mathbb{R}^d$ and $i \in \{1, \dots, d\}$, by

$$X_M^{(i)}(\theta, \mathbf{U}) = -\frac{1}{M} \sum_{r=1}^M \frac{\partial \ell}{\partial \theta_i}(\theta, U_r). \quad (211)$$

The following elementary small-ball criterion is convenient.

Lemma 8.1. *Let $a \in (0, \infty)$, $p \in (0, 1]$, let $\mathcal{V} \subseteq \mathbb{R}^d$, and let (X, U) be an innovation which satisfies for every $\theta \in \mathcal{V}$, $i \in \{1, \dots, d\}$ that*

$$\mathbb{P}(|X^{(i)}(\theta, U)| \geq a) \geq p. \quad (212)$$

Then (X, U) is (c, q) -regular on $\mathcal{V}$ with

$$c = p(1 - e^{-1})a^2 \quad \text{and} \quad q = p(1 - e^{-1}). \quad (213)$$

Proof. Fix $\theta \in \mathcal{V}$, $i \in \{1, \dots, d\}$, and $\lambda \in [0, \infty)$. Assumption (212) gives

$$\mathbb{E}[e^{-\lambda X^{(i)}(\theta, U)^2}] \leq 1 - p + p e^{-\lambda a^2} = 1 - p(1 - e^{-\lambda a^2}). \quad (214)$$

Since $-\ln(1 - x) \geq x$ for $x \in [0, 1)$ and $1 - e^{-x} \geq (1 - e^{-1})(x \wedge 1)$ for $x \in [0, \infty)$, (214) yields

$$\begin{aligned} -\ln \mathbb{E}[e^{-\lambda X^{(i)}(\theta, U)^2}] &\geq p(1 - e^{-\lambda a^2}) \\ &\geq p(1 - e^{-1})(\lambda a^2 \wedge 1) = (c\lambda) \wedge q, \end{aligned} \quad (215)$$

where c and q are given by (213). □

8.2. Strong convexity in the data variable. Our first application uses strong convexity in the data variable. The next estimate combines this property with an upper density bound and will be used to prove Proposition 8.3. The bounded-support assumption provides a uniform bound on the volume of boundary layers of convex sublevel sets. This is needed because an upper density bound alone controls small balls but does not uniformly control boundary layers at arbitrarily high levels.

Lemma 8.2 (Anti-concentration of strongly convex functions). *Let $\mu, B, R \in (0, \infty)$, and suppose that U has a Lebesgue density g on $\mathbb{R}^m$ such that*

$$\|g\|_{L^\infty} \leq B \quad \text{and} \quad \mathbb{P}(|U| < R) = 1. \quad (216)$$

Let $\varphi : \mathbb{R}^m \rightarrow \mathbb{R}$ be μ -strongly convex. Let ω_m denote the Lebesgue volume of the Euclidean unit ball in $\mathbb{R}^m$. For every $\varepsilon \in (0, \infty)$, set

$$\delta_\varepsilon = 2\sqrt{\frac{\varepsilon}{\mu}}. \quad (217)$$

Then, for every $a \in \mathbb{R}$, one has

$$\mathbb{P}(|\varphi(U) - a| \leq \varepsilon) \leq B\omega_m(\delta_\varepsilon^m + (R + 2\delta_\varepsilon)^m - (R + \delta_\varepsilon)^m). \quad (218)$$

Proof. Since a finite-valued strongly convex function on $\mathbb{R}^m$ is continuous and coercive, φ has a unique minimizer u_* . Set $\varphi_* = \varphi(u_*)$. First suppose that $a - \varepsilon < \varphi_*$. If the event in (218) is nonempty, then $a + \varepsilon < \varphi_* + 2\varepsilon$. Strong convexity gives that for all $u \in \mathbb{R}^m$

$$\varphi(u) \geq \varphi_* + \frac{\mu}{2}|u - u_*|^2. \quad (219)$$

It follows from (217) and (219) that

$$\{u \in \mathbb{R}^m : |\varphi(u) - a| \leq \varepsilon\} = \{u \in \mathbb{R}^m : \varphi(u) \leq \varphi_* + 2\varepsilon\} \subseteq B(u_*, \delta_\varepsilon). \quad (220)$$

Therefore, (216) and (220) give

$$\mathbb{P}(|\varphi(U) - a| \leq \varepsilon) \leq \mathbb{P}(U \in B(u_*, \delta_\varepsilon)) \leq B\omega_m\delta_\varepsilon^m. \quad (221)$$

Now suppose that $t = a - \varepsilon \geq \varphi_*$, and set $K_t = \{u \in \mathbb{R}^m : \varphi(u) \leq t\}$. The set K_t is nonempty, closed, and convex. Fix $u \in \mathbb{R}^m \setminus K_t$ and let v be its Euclidean projection onto K_t . Then $\varphi(v) = t$. For every $\alpha \in (0, 1)$, the point $(1 - \alpha)v + \alpha u$ lies outside K_t , and strong convexity yields

$$t < \varphi((1 - \alpha)v + \alpha u) \leq (1 - \alpha)\varphi(v) + \alpha\varphi(u) - \frac{\mu}{2}\alpha(1 - \alpha)|u - v|^2. \quad (222)$$

Using $\varphi(v) = t$ in (222), subtracting t , and dividing by $\alpha > 0$ show that, for every $\alpha \in (0, 1)$,

$$\varphi(u) - t > \frac{\mu}{2}(1 - \alpha)|u - v|^2. \quad (223)$$

Letting α decrease to zero in (223) and recalling that $|u - v| = \text{dist}(u, K_t)$ prove that, for every $u \in \mathbb{R}^m \setminus K_t$,

$$\varphi(u) - t \geq \frac{\mu}{2} \text{dist}(u, K_t)^2. \quad (224)$$

Consequently, (217) and (224) imply, up to a Lebesgue-null boundary,

$$\{u \in \mathbb{R}^m : |\varphi(u) - a| \leq \varepsilon\} \subseteq (K_t + \delta_\varepsilon B(0, 1)) \setminus K_t. \quad (225)$$

Set $L_t = K_t \cap B(0, R + \delta_\varepsilon)$. By (216), only this bounded part of K_t can contribute to the event in (225); more precisely,

$$((K_t + \delta_\varepsilon B(0, 1)) \setminus K_t) \cap B(0, R) \subseteq (L_t + \delta_\varepsilon B(0, 1)) \setminus L_t. \quad (226)$$

Since $L_t \subseteq B(0, R + \delta_\varepsilon)$, the Steiner formula and monotonicity of the intrinsic volumes, together with (226), yield

$$|((K_t + \delta_\varepsilon B(0, 1)) \setminus K_t) \cap B(0, R)| \leq \omega_m((R + 2\delta_\varepsilon)^m - (R + \delta_\varepsilon)^m). \quad (227)$$

Equations (216), (225), and (227) prove (218). $\square$

Proposition 8.3 (Strongly convex coordinate gradients). *Let $m, d, M \in \mathbb{N}$, $\mu, B, R \in (0, \infty)$, and let $\mathcal{V} \subseteq \mathbb{R}^d$. Let U be an $\mathbb{R}^m$ -valued random variable with a Lebesgue density g satisfying*

$$\|g\|_{L^\infty} \leq B, \quad \text{and} \quad \mathbb{P}(|U| < R) = 1. \quad (228)$$

Let $U_1, \dots, U_M$ be independent copies of U , and set $\mathbf{U} = (U_1, \dots, U_M)$. Let $\ell : \mathbb{R}^d \times \mathbb{R}^m \rightarrow \mathbb{R}$ be differentiable in its first argument. For every $\theta \in \mathcal{V}$, $i \in \{1, \dots, d\}$, define

$$\varphi_{\theta,i} : \mathbb{R}^m \rightarrow \mathbb{R}, \quad \varphi_{\theta,i}(u) = \frac{\partial \ell}{\partial \theta_i}(\theta, u), \quad (229)$$

and assume that $\varphi_{\theta,i}$ is μ -strongly convex. For every $\theta \in \mathcal{V}$, $i \in \{1, \dots, d\}$, define

$$X_M^{(i)}(\theta, \mathbf{U}) = -\frac{1}{M} \sum_{r=1}^M \frac{\partial \ell}{\partial \theta_i}(\theta, U_r). \quad (230)$$

Let ω_m denote the Lebesgue volume of the Euclidean unit ball in $\mathbb{R}^m$, and set

$$C_m = 1 + m3^{m-1} \quad (231)$$

and

$$s_0 = \mu \min \left\{ \frac{R^2}{4}, \frac{1}{16B^2\omega_m^2 C_m^2 R^{2m-2}} \right\}. \quad (232)$$

Then the minibatch innovation $(X_M, \mathbf{U})$ is (c_M, q) -regular on $\mathcal{V}$ with

$$c_M = \frac{(1 - e^{-1})s_0^2}{2M^2} \quad \text{and} \quad q = \frac{1 - e^{-1}}{2}. \quad (233)$$

If, in addition, the coordinate derivatives of ℓ are uniformly bounded on $\mathcal{V} \times \{u \in \mathbb{R}^m : |u| < R\}$, then assumption (9) in [Theorem 1.1](#) is satisfied for X_M .

Proof. For every $s \in (0, s_0]$, set $\delta_s = 2\sqrt{\frac{s}{\mu}}$. Equation (232) then gives $\delta_s \leq R$. Hence

$$\begin{aligned} \delta_s^m + (R + 2\delta_s)^m - (R + \delta_s)^m &\leq R^{m-1}\delta_s + m(R + 2\delta_s)^{m-1}\delta_s \\ &\leq C_m R^{m-1}\delta_s. \end{aligned} \quad (234)$$

By (228), [Lemma 8.2](#) applies to every $\varphi_{\theta,i}$. Equations (218), (232), and (234) give

$$\sup_{\substack{\theta \in \mathcal{V}, 1 \leq i \leq d \\ a \in \mathbb{R}}} \mathbb{P}(|\varphi_{\theta,i}(U) - a| \leq s_0) \leq B\omega_m C_m R^{m-1}\delta_{s_0} = 2B\omega_m C_m R^{m-1} \sqrt{\frac{s_0}{\mu}} \leq \frac{1}{2}. \quad (235)$$

Fix $\theta \in \mathcal{V}$ and $i \in \{1, \dots, d\}$. By (230) and (229), conditional on $U_2, \dots, U_M$, the event

$$\left\{ \left| X_M^{(i)}(\theta, \mathbf{U}) \right| \leq \frac{s_0}{M} \right\} \quad (236)$$

is an interval event of width $2s_0$ for $\varphi_{\theta,i}(U_1)$, whose center may depend on $U_2, \dots, U_M$. Therefore, (235) and (236) imply

$$\mathbb{P}\left(\left| X_M^{(i)}(\theta, \mathbf{U}) \right| \geq \frac{s_0}{M}\right) \geq \frac{1}{2}. \quad (237)$$

[Lemma 8.1](#), applied with $a = s_0/M$ and $p = 1/2$, proves the asserted regularity constants. The boundedness assertion follows directly from (230). $\square$

The preceding criterion applies naturally to risk-sensitive objectives in which the parameter changes the exponential weighting of a random energy. Unlike feature-mean estimation, the resulting optimizer depends on the Laplace transform of the energy and hence on more than one fixed moment.

Example 8.4 (Risk-sensitive exponential tilting). Let $\bar{\theta}, \mu, B, R \in (0, \infty)$, assume (216), and let $\mathcal{V} \subseteq [0, \bar{\theta}]^d$ be nonempty and compact. Let $G: \mathbb{R}^d \rightarrow \mathbb{R}$ be continuously differentiable. For every $i \in \{1, \dots, d\}$, let $\varphi_i: \mathbb{R}^m \rightarrow [0, \infty)$ be twice continuously differentiable and μ -strongly convex. Consider the risk-sensitive loss

$$\ell_{\text{risk}}(\theta, u) = G(\theta) + \sum_{i=1}^d e^{\theta_i \varphi_i(u)}. \quad (238)$$

For every $\theta \in \mathcal{V}$, $i \in \{1, \dots, d\}$, $u \in \mathbb{R}^m$, one has

$$\frac{\partial \ell_{\text{risk}}}{\partial \theta_i}(\theta, u) = \frac{\partial G}{\partial \theta_i}(\theta) + \varphi_i(u) e^{\theta_i \varphi_i(u)}. \quad (239)$$

For symmetric matrices $A, B \in \mathbb{R}^{m \times m}$, write $A \geq B$ if $A - B$ is positive semidefinite. Moreover, for every $\theta \in \mathcal{V}$, $i \in \{1, \dots, d\}$, $u \in \mathbb{R}^m$, one has

$$\begin{aligned} \text{Hess}_u \left(\varphi_i(u) e^{\theta_i \varphi_i(u)} \right) &= e^{\theta_i \varphi_i(u)} (1 + \theta_i \varphi_i(u)) \text{Hess } \varphi_i(u) \\ &\quad + \theta_i e^{\theta_i \varphi_i(u)} (2 + \theta_i \varphi_i(u)) \nabla \varphi_i(u) \nabla \varphi_i(u)^\top \\ &\geq \mu I_m. \end{aligned} \quad (240)$$

Indeed, $\theta_i, \varphi_i(u) \geq 0$ and $\text{Hess } \varphi_i(u) \geq \mu I_m$. Equations (239) and (240) show that every coordinate gradient of (238) is μ -strongly convex as a function of u . Therefore, Proposition 8.3 applies. In particular, with s_0 defined in (232), the minibatch innovation driven by (238) is (c_M, q) -regular on $\mathcal{V}$, where c_M and q are given in (233).

The continuity assumptions and the compactness of $\mathcal{V} \times B(0, R)$ also imply that the coordinate gradients in (239) are uniformly bounded on this set. Hence assumption (9) in Theorem 1.1 is satisfied by the resulting minibatch innovation.

For every $\theta \in \mathcal{V}$, the associated population objective is given by

$$F_{\text{risk}}(\theta) = G(\theta) + \sum_{i=1}^d \mathbb{E} \left[e^{\theta_i \varphi_i(U)} \right]. \quad (241)$$

The gradient of F_{risk} is the negative of the mean innovation. For example, if $G(\theta) = -\sum_{i=1}^d b_i \theta_i$ for constants $b_1, \dots, b_d \in \mathbb{R}$, every interior critical point satisfies, for every $i \in \{1, \dots, d\}$,

$$\mathbb{E} \left[\varphi_i(U) e^{\theta_i \varphi_i(U)} \right] = b_i. \quad (242)$$

Thus the optimized parameters are exponential-tilting parameters determined by the Laplace transforms of the random energies $\varphi_i(U)$.

8.3. Nonlinear regression. We finally consider nonlinear regression with vector-valued responses. Let $k \in \mathbb{N}$, let $\mathcal{Z} \subseteq \mathbb{R}^k$ be a Borel set, let $U = (Z, Y)$ take values in $\mathcal{Z} \times \mathbb{R}^m$, and let $(\Gamma^\theta; \theta \in \mathbb{R}^d)$ be a family of measurable maps from $\mathcal{Z}$ to $\mathbb{R}^m$ that is differentiable with respect to θ . For every $\theta \in \mathbb{R}^d$, $z \in \mathcal{Z}$, $y \in \mathbb{R}^m$, and $i \in \{1, \dots, d\}$, the quadratic loss and its associated coordinate innovation are given by

$$\begin{aligned} \ell_{\text{reg}}(\theta, (z, y)) &= \frac{1}{2} \left| \Gamma^\theta(z) - y \right|^2, \\ X^{(i)}(\theta, (z, y)) &= - \left\langle \frac{\partial \Gamma^\theta}{\partial \theta_i}(z), \Gamma^\theta(z) - y \right\rangle. \end{aligned} \quad (243)$$

The following result first bounds negative moments of the scalar residual in the direction selected by the derivative of the regression function. A reverse Hölder inequality turns this into a uniform positive lower bound for the first absolute moment. Compactness provides a uniform upper bound for the innovation, and the two moment bounds then yield regularity.

Proposition 8.5 (Compactly supported nonlinear regression). *Let $\mathcal{V} \subseteq \mathbb{R}^d$ be nonempty and compact, let $r \in (0, 1/2)$, $\delta \in (0, \infty)$, and suppose that there exist nonempty compact sets $K_Z \subseteq \mathcal{Z}$ and $K_Y \subseteq \mathbb{R}^m$ and a constant $B \in (0, \infty)$ with the following properties:*

(i) *The random variables are compactly supported in the sense that*

$$\mathbb{P}(Z \in K_Z, Y \in K_Y) = 1. \quad (244)$$

(ii) *For $\mathbb{P}_Z$ -almost every $z \in K_Z$, the conditional law of Y given $Z = z$ has a Lebesgue density p_z satisfying*

$$\|p_z\|_{L^\infty} \leq B. \quad (245)$$

(iii) *The map $(\theta, z) \mapsto \Gamma^\theta(z)$ and, for every $i \in \{1, \dots, d\}$, the map $(\theta, z) \mapsto \frac{\partial \Gamma^\theta}{\partial \theta_i}(z)$ are continuous on $\mathcal{V} \times K_Z$.*

(iv) *The coordinate derivatives satisfy the uniform moment lower bound*

$$D_r = \inf_{\theta \in \mathcal{V}} \inf_{1 \leq i \leq d} \mathbb{E} \left[\left| \frac{\partial \Gamma^\theta}{\partial \theta_i}(Z) \right|^r \right] \geq \delta. \quad (246)$$

Choose $R \in (0, \infty)$ such that $K_Y \subseteq B(0, R)$, let ω_{m-1} denote the volume of the unit ball in $\mathbb{R}^{m-1}$, use the convention $\omega_0 = 1$, and set

$$\begin{aligned} \mu_r &= \frac{\delta^{1/r}}{2B\omega_{m-1}R^{m-1}} \left(\frac{1-2r}{1-r} \right)^{(1-r)/r}, \\ C_X &= \max_{\theta \in \mathcal{V}, \substack{z \in K_Z, y \in K_Y \\ 1 \leq i \leq d}} \left\| \frac{\partial \Gamma^\theta}{\partial \theta_i}(z), \Gamma^\theta(z) - y \right\|. \end{aligned} \quad (247)$$

Then $\mu_r, C_X \in (0, \infty)$. The innovation (X, U) defined in (243) is (c_r, q_r) -regular on $\mathcal{V}$ with

$$c_r = \frac{(1-e^{-1})\mu_r^3}{8C_X}, \quad q_r = \frac{(1-e^{-1})\mu_r}{2C_X}. \quad (248)$$

Proof. Since $r \in (0, 1/2)$, one has $r/(1-r) \in (0, 1)$. We first prove the uniform first-moment lower bound that drives the argument. Fix $\theta \in \mathcal{V}$ and $i \in \{1, \dots, d\}$. In every normalized derivative below, the quotient is understood as an arbitrary fixed unit vector in $\mathbb{R}^m$ whenever $\partial \Gamma^\theta / \partial \theta_i$ vanishes. With this convention the normalized derivative is measurable, has unit norm, and its product with $|\partial \Gamma^\theta / \partial \theta_i|$ remains the original derivative.

By (244), the conditional density p_z vanishes almost everywhere outside K_Y for $\mathbb{P}_Z$ -almost every $z \in K_Z$. Fix such a z . After an orthogonal change of coordinates, Fubini's theorem shows that the conditional density of the residual projected onto the normalized direction specified above is bounded by $B\omega_{m-1}R^{m-1}$: the density p_z is bounded by B according to (245), and every hyperplane slice of $K_Y \subseteq B(0, R)$ has $(m-1)$ -dimensional Hausdorff measure at most $\omega_{m-1}R^{m-1}$. Consequently, for every $h \in (0, \infty)$,

$$\mathbb{P} \left(\left\| \frac{\partial \Gamma^\theta / \partial \theta_i(z)}{|\partial \Gamma^\theta / \partial \theta_i(z)|}, \Gamma^\theta(z) - Y \right\| \leq h \mid Z = z \right) \leq 2B\omega_{m-1}R^{m-1}h. \quad (249)$$

Letting h decrease to zero in (249) shows that the projected residual is conditionally nonzero almost surely. For the fixed θ, i , and z , denote its absolute value by W and set $C_{\text{pr}} = 2B\omega_{m-1}R^{m-1}$. We get with (249) that

$$\begin{aligned} \mathbb{E} \left[\left\| \frac{\partial \Gamma^\theta / \partial \theta_i(z)}{|\partial \Gamma^\theta / \partial \theta_i(z)|}, \Gamma^\theta(z) - Y \right\|^{-r/(1-r)} \mid Z = z \right] &= \int_0^\infty \mathbb{P}(W^{-r/(1-r)} \geq t \mid Z = z) dt \\ &= \int_0^\infty \mathbb{P}(W \leq t^{-(1-r)/r} \mid Z = z) dt \leq \int_0^{C_{\text{pr}}^{r/(1-r)}} 1 dt + C_{\text{pr}} \int_{C_{\text{pr}}^{r/(1-r)}}^\infty t^{-(1-r)/r} dt \\ &= C_{\text{pr}}^{r/(1-r)} + \frac{r}{1-2r} C_{\text{pr}}^{r/(1-r)} = \frac{1-r}{1-2r} (2B\omega_{m-1}R^{m-1})^{r/(1-r)}. \end{aligned} \quad (250)$$

Taking expectations in (250) and using (244) give

$$\mathbb{E} \left[\left\| \frac{\partial \Gamma^\theta / \partial \theta_i(Z)}{|\partial \Gamma^\theta / \partial \theta_i(Z)|}, \Gamma^\theta(Z) - Y \right\|^{-r/(1-r)} \right] \leq \frac{1-r}{1-2r} (2B\omega_{m-1}R^{m-1})^{r/(1-r)}. \quad (251)$$

The reverse Hölder inequality used below follows from the ordinary Hölder inequality with conjugate exponents $1/r$ and $1/(1-r)$. Equations (243), (246), and (251) therefore yield

$$\begin{aligned} \mathbb{E}[|X^{(i)}(\theta, U)|] &= \mathbb{E} \left[\left\| \frac{\partial \Gamma^\theta}{\partial \theta_i}(Z) \right\| \left\| \frac{\partial \Gamma^\theta / \partial \theta_i(Z)}{|\partial \Gamma^\theta / \partial \theta_i(Z)|}, \Gamma^\theta(Z) - Y \right\| \right] \\ &\geq \mathbb{E} \left[\left\| \frac{\partial \Gamma^\theta}{\partial \theta_i}(Z) \right\|^r \right]^{1/r} \mathbb{E} \left[\left\| \frac{\partial \Gamma^\theta / \partial \theta_i(Z)}{|\partial \Gamma^\theta / \partial \theta_i(Z)|}, \Gamma^\theta(Z) - Y \right\|^{-r/(1-r)} \right]^{-(1-r)/r} \\ &\geq \delta^{1/r} \left[\frac{1-r}{1-2r} (2B\omega_{m-1}R^{m-1})^{r/(1-r)} \right]^{-(1-r)/r} = \mu_r, \end{aligned} \quad (252)$$

where the last identity follows from (247). Since θ and i were arbitrary, (252) holds uniformly in $\theta \in \mathcal{V}$ and $i \in \{1, \dots, d\}$.

The continuity assumptions and the compactness of $\mathcal{V} \times K_Z \times K_Y$ imply that the maximum defining C_X in (247) exists and is finite. Moreover, by (244) and (243), for every $\theta \in \mathcal{V}$ and $i \in \{1, \dots, d\}$ one has, almost surely,

$$|X^{(i)}(\theta, U)| \leq C_X. \quad (253)$$

Equations (253) and (252) first show that $C_X \geq \mu_r > 0$. They also imply that, for every $\theta \in \mathcal{V}$ and $i \in \{1, \dots, d\}$,

$$\mu_r \leq \mathbb{E}[|X^{(i)}(\theta, U)|] \leq \frac{\mu_r}{2} + C_X \mathbb{P}\left(|X^{(i)}(\theta, U)| \geq \frac{\mu_r}{2}\right). \quad (254)$$

It follows from (254) that, for every $\theta \in \mathcal{V}$ and $i \in \{1, \dots, d\}$,

$$\mathbb{P}\left(|X^{(i)}(\theta, U)| \geq \frac{\mu_r}{2}\right) \geq \frac{\mu_r}{2C_X}. \quad (255)$$

Since $\mu_r/2 \in (0, \infty)$ and $\mu_r/(2C_X) \in (0, 1]$, equation (255) and Lemma 8.1 apply with threshold $\mu_r/2$ and probability $\mu_r/(2C_X)$. The constants in (213) are then

$$\frac{\mu_r}{2C_X}(1 - e^{-1})\left(\frac{\mu_r}{2}\right)^2 = \frac{(1 - e^{-1})\mu_r^3}{8C_X} = c_r \quad \text{and} \quad \frac{\mu_r}{2C_X}(1 - e^{-1}) = \frac{(1 - e^{-1})\mu_r}{2C_X} = q_r, \quad (256)$$

where the last identities follow from (248). This proves the claim. $\square$

Proof of Theorem 1.2. Minibatch innovation and regularity. For every $n \in \mathbb{N}$, set $\mathbf{Y}_n = (Y_{n,1}, \dots, Y_{n,M})$ and, for every $i \in \{1, \dots, d\}$, set

$$G_n^{(i)} = \frac{1}{M} \sum_{r=1}^M (\nabla_{\theta_i} L)(\theta_{n-1}, Y_{n,r}). \quad (257)$$

For $\mathbf{y} = (y_1, \dots, y_M) \in (\mathbb{R}^m)^M$, set $\bar{y}_M = M^{-1} \sum_{r=1}^M y_r$ and define the innovation

$$X_M(\theta, \mathbf{y}) = -\frac{1}{M} \sum_{r=1}^M \nabla_{\theta} L(\theta, y_r) = 2A^\top(\bar{y}_M - A\theta). \quad (258)$$

If $v_n = w_n/(1 - \beta^n)$ for $n \in \mathbb{N}$ and $v_0 = 0$, then (16) is precisely the RMSprop algorithm driven by $(X_M, \mathbf{Y}_1)$ with parameter tuple $(\beta, \varepsilon, (\gamma_n)_{n \in \mathbb{N}}, \theta_0)$.

Let p denote a bounded Lebesgue density of $Y_{1,1}$. The average $\bar{Y}_{1,M} = M^{-1} \sum_{r=1}^M Y_{1,r}$ is supported by $\text{conv}(K)$. Its density is

$$p_M(y) = M^m p^{*M}(My). \quad (259)$$

Young's convolution inequality gives $\|p_M\|_{L^\infty} \leq M^m \|p\|_{L^\infty} < \infty$. Fix $r \in (0, 1/2)$ and set

$$\begin{aligned} V_\kappa &= \{\theta \in \mathbb{R}^d : |\theta| \leq \kappa\}, & K_Y &= \text{conv}(K), & B &= M^m \|p\|_{L^\infty}, \\ \delta &= \min_{1 \leq i \leq d} |Ae_i|^r > 0. \end{aligned} \quad (260)$$

Apply Proposition 8.5 with $\mathcal{Z} = K_Z = \{0\} \subseteq \mathbb{R}$, $Z = 0$, $\Gamma^\theta(0) = A\theta$, response $\bar{Y}_{1,M}$, and the parameters in (260). Its derivative condition holds because

$$\inf_{\substack{\theta \in V_\kappa \\ 1 \leq i \leq d}} \mathbb{E} \left[\left| \frac{\partial \Gamma^\theta}{\partial \theta_i}(Z) \right|^r \right] = \delta. \quad (261)$$

Proposition 8.5 and (261) establish regularity and boundedness for the innovation in (258) without the factor 2. Scaling by 2 changes regularity parameters (c, q) into $(4c, q)$ and doubles the uniform bound. Hence there exist $c_{\text{lin}}, q_{\text{lin}}, C_X \in (0, \infty)$ such that $(X_M, \mathbf{Y}_1)$ is $(c_{\text{lin}}, q_{\text{lin}})$ -regular on V_κ and, for every $\theta \in V_\kappa$ and $i \in \{1, \dots, d\}$,

$$|X_M^{(i)}(\theta, \mathbf{Y}_1)| \leq C_X \quad \text{almost surely.} \quad (262)$$

Since A has no zero column, it is nonzero. Define λ_* as the smallest positive eigenvalue of $A^\top A$. Since an upper bound remains valid when its constant is increased, we may and do assume that

$$C_X \geq 4\lambda_*\Gamma. \quad (263)$$

Choose $\beta_0 \in (\max\{1/2, e^{-q_{\text{lin}}/3}\}, 1)$ and set $\rho = q_{\text{lin}} + 3 \ln \beta_0 > 0$.

Objective geometry and uniform constants. Let $\theta_* \in \mathbb{R}^d$ solve the normal equation $A^\top A \theta_* = A^\top \mathbb{E}[Y_{1,1}]$. Such a solution exists because $\text{range}(A^\top) = \text{range}(A^\top A)$, and it minimizes the population risk. Then

$$F(\theta) = |A(\theta - \theta_*)|^2. \quad (264)$$

Equations (258) and (264) show that $f_{X_M} = -\nabla F$ and, by the spectral theorem, that, for every $\theta \in \mathbb{R}^d$,

$$F(\theta) \leq \frac{1}{4\lambda_*} |f_{X_M}(\theta)|^2. \quad (265)$$

Moreover, ∇F is globally $2\|A^\top A\|$ -Lipschitz continuous. Set

$$C_{\text{Loj}} = \frac{1}{4\lambda_*}, \quad \eta = \frac{1}{2C_{\text{Loj}}(C_X + 1)}. \quad (266)$$

Thus η depends only on the fixed problem data. Now fix $\mathfrak{C} \in (0, \infty)$, set $C_{\text{dec}} = \mathfrak{C}$, and let $C_0, C_1 \in (0, \infty)$ be the constants furnished by [Theorem 1.1](#) for

$$c = c_{\text{lin}}, \quad q = q_{\text{lin}}, \quad L_f = 2\|A^\top A\|, \quad (267)$$

and the fixed values of $\rho, C_X, C_{\text{Loj}}, C_{\text{dec}}$. In particular, C_0 and C_1 are independent of θ_0 , the particular step-size sequence, β , and ε .

Fix such a step-size sequence, $\beta \in [1/2, 1)$, $\varepsilon \in [0, 1]$, and $\theta_0 \in \mathbb{R}^d$, and let $(\theta_n)_{n \in \mathbb{N}_0}$ be the process in the statement. Set $\tau = \inf\{j \in \mathbb{N}_0 : |\theta_j| > \kappa\}$ and, for every $n \in \mathbb{N}_0$, set

$$\mathcal{I}_n = \{\tau \geq n\} = \bigcap_{j=0}^{n-1} \{|\theta_j| \leq \kappa\}. \quad (268)$$

Here $\mathcal{I}_0 = \Omega$ by the empty-intersection convention.

The case $n = 1$ is immediate. Indeed, by (257), $w_1^{(i)} = (1 - \beta)|G_1^{(i)}|^2$ and (5) give $|\theta_1 - \theta_0| \leq \sqrt{d}\gamma_1$. Thus (264) and the triangle inequality give

$$F(\theta_1)^{1/2} \leq F(\theta_0)^{1/2} + \|A^\top A\|^{1/2} |\theta_1 - \theta_0| \leq F(\theta_0)^{1/2} + \sqrt{d\|A^\top A\|} \gamma_1. \quad (269)$$

Consequently, (269) yields

$$\mathbb{E}[F(\theta_1)\mathbb{1}_{\mathcal{I}_1}] \leq \left(F(\theta_0)^{1/2} + \sqrt{d\|A^\top A\|} \gamma_1\right)^2. \quad (270)$$

The regime $\beta \in [\beta_0, 1)$. For every $n \geq 2$, the range of the step sizes and (263) give

$$\gamma_n \leq \Gamma \leq C_{\text{Loj}} C_X \leq C_{\text{Loj}}(C_X + \varepsilon). \quad (271)$$

Moreover, the assumed step-size comparison implies, for every $j, n \in \mathbb{N}$ with $j \leq n$, that

$$\gamma_j \leq C_{\text{dec}} \gamma_n \exp\left(\frac{\eta}{2} \sum_{k=j+1}^n \gamma_k\right). \quad (272)$$

If $\beta \in [\beta_0, 1)$, the assumptions of [Theorem 1.1](#) follow from (262), (265), the preceding step-size bounds, and (271); moreover, $q_{\text{lin}} + 3 \ln \beta \geq \rho$. Thus [Theorem 1.1](#) and (268) show that, for every $n \geq 2$,

$$\mathbb{E}[F(\theta_n)\mathbb{1}_{\mathcal{I}_n}] \leq \left(F(\theta_0)^{1/2} + \sqrt{d\|A^\top A\|} \gamma_1\right)^2 \exp\left(-\eta \sum_{k=2}^n \gamma_k\right) + C_0 \gamma_n + C_1 (1 - \beta)^2. \quad (273)$$

The regime $\beta \in [1/2, \beta_0]$. The recursion for $w_n^{(i)}$ and (257) imply $w_n^{(i)} \geq (1 - \beta)|G_n^{(i)}|^2$. Hence, for every $n \in \mathbb{N}$ and $i \in \{1, \dots, d\}$,

$$\left| \left[\varepsilon + (1 - \beta^n)^{-1/2} (w_n^{(i)})^{1/2} \right]^{-1} G_n^{(i)} \right| \leq \sqrt{\frac{1 - \beta^n}{1 - \beta}} \leq \frac{1}{\sqrt{1 - \beta_0}}. \quad (274)$$

The first inequality in (274) also holds when its denominator vanishes, by the convention in (5).

Set

$$R = \kappa + \frac{\sqrt{d}\mathfrak{C}}{\sqrt{1 - \beta_0}}, \quad C_F = \sup_{|\theta| \leq R} F(\theta) < \infty. \quad (275)$$

For every $n \geq 2$, one has $|\theta_{n-1}| \leq \kappa$ on $\mathcal{I}_n$. Consequently, the bound $\gamma_n \leq \mathfrak{C}$ and (274) imply

$$|\theta_n| \leq |\theta_{n-1}| + \frac{\sqrt{d}\gamma_n}{\sqrt{1 - \beta_0}} \leq R \quad \text{on } \mathcal{I}_n. \quad (276)$$

Since $(1 - \beta)^2 \geq (1 - \beta_0)^2$, it follows from (276) that, for every $n \geq 2$,

$$\mathbb{E}[F(\theta_n)\mathbb{1}_{\mathcal{I}_n}] \leq C_F \leq \frac{C_F}{(1 - \beta_0)^2} (1 - \beta)^2. \quad (277)$$

Conclusion. The first term on the right-hand side of (17) is nonnegative. Therefore, (270), (273), and (277) prove (17) for every $n \in \mathbb{N}$ and $\beta \in [1/2, 1)$ with

$$c = \max \left\{ C_0, C_1, \frac{C_F}{(1 - \beta_0)^2} \right\}. \quad (278)$$

This constant may depend on $\mathfrak{C}$, but is independent of θ_0 , the particular step-size sequence, β , and ε , as required. $\square$

Acknowledgements. This work has been partially funded by the National Science Foundation of China (NSFC) under grant number W2531010. This work has also been partially funded by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) under Germany's Excellence Strategy EXC 2044-390685587, Mathematics Münster: Dynamics-Geometry-Structure.

REFERENCES

- [1] Anas Barakat and Pascal Bianchi. Convergence and dynamical behavior of the Adam algorithm for nonconvex stochastic optimization. *SIAM J. Optim.*, 31(1):244–274, 2021.
- [2] Bilel Bensaid, Gaël Poëtte, and Rodolphe Turpault. Convergence of the Iterates for Momentum and RMSProp for Local Smooth Functions: Adaptation is the Key. *arXiv:2407.15471*, 2024.
- [3] Xiangyi Chen, Sijia Liu, Ruoyu Sun, and Mingyi Hong. On the Convergence of A Class of Adam-Type Algorithms for Non-Convex Optimization. In *International Conference on Learning Representations (ICLR)*, 2019.
- [4] Soham De, Anirbit Mukherjee, and Enayat Ullah. Convergence guarantees for RMSProp and ADAM in non-convex optimization and an empirical comparison to Nesterov acceleration. *arXiv:1807.06766*, 2018.
- [5] Alexandre Défossez, Leon Bottou, Francis Bach, and Nicolas Usunier. A Simple Convergence Proof of Adam and Adagrad. *Transactions on Machine Learning Research*, 2022.
- [6] Steffen Dereich, Thang Do, and Arnulf Jentzen. Uniform a priori bounds and error analysis for the Adam stochastic gradient descent optimization method. *arXiv:2603.18899*, 2026.
- [7] Steffen Dereich, Thang Do, Arnulf Jentzen, and Philippe von Wurstemberger. Adam symmetry theorem: characterization of the convergence of the stochastic Adam optimizer. *arXiv:2511.06675*, 2025.
- [8] Steffen Dereich, Robin Graeber, and Arnulf Jentzen. Non-convergence of Adam and other adaptive stochastic gradient descent optimization methods for non-vanishing learning rates. *arXiv:2407.08100*, 2024.
- [9] Steffen Dereich, Robin Graeber, Arnulf Jentzen, and Adrian Riekert. Asymptotic stability properties and a priori bounds for Adam and other gradient descent optimization methods. *arXiv:2509.10476*, 2026.
- [10] Steffen Dereich and Arnulf Jentzen. Convergence rates for the Adam optimizer. *arXiv:2407.21078*, 2024.
- [11] Steffen Dereich, Arnulf Jentzen, and Adrian Riekert. Sharp higher order convergence rates for the Adam optimizer. *arXiv:2504.19426*, 2025.

- [12] Naum Dimitrieski, Maria Christine Honecker, Carsten Scherer, and Christian Ebenbauer. Global Stability and Step Size Robustness of RMSProp. [arXiv:2603.15823](#), 2026.
- [13] Thang Do, Sonja Hannibal, and Arnulf Jentzen. Non-convergence to global minimizers in data driven supervised deep learning: Adam and stochastic gradient descent optimization provably fail to converge to global minimizers in the training of deep neural networks with ReLU activation. *J. Math. Anal. Appl.*, 564(2):Paper No. 130724, 2026.
- [14] Thang Do, Arnulf Jentzen, and Adrian Riekert. Non-convergence to the optimal risk for Adam and stochastic gradient descent optimization in the training of deep neural networks. [arXiv:2503.01660](#), 2025.
- [15] John Duchi, Elad Hazan, and Yoram Singer. Adaptive subgradient methods for online learning and stochastic optimization. *J. Mach. Learn. Res.*, 12:2121–2159, 2011.
- [16] Sébastien Gadat and Ioana Gavra. Asymptotic study of stochastic adaptive algorithms in non-convex landscape. *J. Mach. Learn. Res.*, 23:Paper No. [228], 54, 2022.
- [17] Antoine Godichon-Baggioni and Pierre Tarrago. Non asymptotic analysis of Adaptive stochastic gradient algorithms and applications. [arXiv:2303.01370](#), 2023.
- [18] Steven Heilman and Sampad Mohanty. On the Convergence of Adam, Revisited. [arXiv:2607.03519](#), 2026.
- [19] Yusu Hong and Junhong Lin. On Convergence of Adam for Stochastic Optimization under Relaxed Assumptions. In *Advances in Neural Information Processing Systems*, volume 37, pages 10827–10877, 2024.
- [20] Shokhrukh Ibragimov and Arnulf Jentzen. Unified convergence analysis for gradient descent optimization methods in the training of deep neural networks. [arXiv:2607.04233](#), 2026.
- [21] Arnulf Jentzen, Benno Kuckuck, and Philippe von Wurstemberger. Mathematical Introduction to Deep Learning: Methods, Implementations, and Theory. [arXiv:2310.20360](#), 2023.
- [22] Arnulf Jentzen and Adrian Riekert. Non-convergence to global minimizers for Adam and stochastic gradient descent optimization and constructions of local minimizers in the training of artificial neural networks. *SIAM/ASA J. Uncertain. Quantif.*, 13(3):1294–1333, 2025.
- [23] Arnulf Jentzen and Philippe von Wurstemberger. Lower error bounds for the stochastic gradient descent optimization algorithm: sharp convergence rates for slowly and fast decaying learning rates. *J. Complexity*, 57:101438, 16, 2020.
- [24] Ruinan Jin and Xiaoyu Wang. Asymptotic Convergence and Stability of Adaptive Gradient Methods in Smooth Non-convex Optimization. [arXiv:2601.01853](#), 2026.
- [25] Diederik P. Kingma and Jimmy Ba. Adam: A Method for Stochastic Optimization. [arXiv:1412.6980](#), 2014.
- [26] Haochuan Li, Alexander Rakhlin, and Ali Jadbabaie. Convergence of Adam Under Relaxed Assumptions. [arXiv:2304.13972](#), 2024.
- [27] Jinlan Liu, Dongpo Xu, Huisheng Zhang, and Danilo Mandic. On hyper-parameter selection for guaranteed convergence of RMSProp. *Cognitive Neurodynamics*, 18(6):3227–3237, 2024.
- [28] Ilya Loshchilov and Frank Hutter. Decoupled Weight Decay Regularization. [arXiv:1711.05101](#), 2017.
- [29] Chao Ma, Lei Wu, and Weinan E. A Qualitative Study of the Dynamic Behavior for Adaptive Gradient Algorithms. [arXiv:2009.06125](#), 2020.
- [30] Sadhika Malladi, Kaifeng Lyu, Abhishek Panigrahi, and Sanjeev Arora. On the SDEs and Scaling Rules for Adaptive Gradient Algorithms. [arXiv:2205.10287](#), 2022.
- [31] Mahesh Chandra Mukkamala and Matthias Hein. Variants of RMSProp and Adagrad with Logarithmic Regret Bounds. [arXiv:1706.05507](#), 2017.
- [32] Sashank J. Reddi, Satyen Kale, and Sanjiv Kumar. On the Convergence of Adam and Beyond. [arXiv:1904.09237](#), 2019.
- [33] Sebastian Ruder. An overview of gradient descent optimization algorithms. [arXiv:1609.04747](#), 2016.
- [34] Naichen Shi, Dawei Li, Mingyi Hong, and Ruoyu Sun. RMSprop Converges with Proper Hyper-Parameter. In *International Conference on Learning Representations (ICLR)*, 2021.
- [35] Tijmen Tieleman and Geoffrey Hinton. Lecture 6.5—RMSProp: Divide the gradient by a running average of its recent magnitude. *COURSERA: Neural Networks for Machine Learning*, 4(2):26–31, 2012.
- [36] Bohan Wang, Jingwen Fu, Huishuai Zhang, Nanning Zheng, and Wei Chen. Closing the Gap Between the Upper Bound and the Lower Bound of Adam’s Iteration Complexity. [arXiv:2310.17998](#), 2023.
- [37] Dongpo Xu, Shengdong Zhang, Huisheng Zhang, and Danilo P. Mandic. Convergence of the RMSProp deep learning method with penalty for nonconvex optimization. *Neural Networks*, 139:17–23, 2021.
- [38] Yaxin Yu, Long Chen, and Minfu Feng. Adam-SHANG: A Convergent Adam-Type Method for Stochastic Smooth Convex Optimization. [arXiv:2605.12878](#), 2026.
- [39] Manzil Zaheer, Sashank Reddi, Devendra Sachan, Satyen Kale, and Sanjiv Kumar. Adaptive Methods for Nonconvex Optimization. In *Advances in Neural Information Processing Systems*, volume 31, 2018.
- [40] Qi Zhang, Yi Zhou, and Shaofeng Zou. Convergence Guarantees for RMSProp and Adam in Generalized-smooth Non-convex Optimization with Affine Noise Variance. *Transactions on Machine Learning Research*, 2025.
- [41] Yushun Zhang, Congliang Chen, Naichen Shi, Ruoyu Sun, and Zhi-Quan Luo. Adam Can Converge Without Any Modification On Update Rules. [arXiv:2208.09632](#), 2022.

- [42] Dongruo Zhou, Jinghui Chen, Yuan Cao, Ziyang Yang, and Quanquan Gu. On the Convergence of Adaptive Gradient Methods for Nonconvex Optimization. *Transactions on Machine Learning Research*, 2024.
- [43] Fangyu Zou, Li Shen, Zequn Jie, Weizhong Zhang, and Wei Liu. A Sufficient Condition for Convergences of Adam and RMSProp. *arXiv:1811.09358*, 2018.

STEFFEN DEREICH, INSTITUTE FOR MATHEMATICAL STOCHASTICS, FACULTY OF MATHEMATICS AND COMPUTER SCIENCE, UNIVERSITY OF MÜNSTER, GERMANY

Email address: `steffen.dereich@uni-muenster.de`

ARNULF JENTZEN, SCHOOL OF DATA SCIENCE AND SCHOOL OF ARTIFICIAL INTELLIGENCE, THE CHINESE UNIVERSITY OF HONG KONG, SHENZHEN (CUHK-SHENZHEN), CHINA & APPLIED MATHEMATICS: INSTITUTE FOR ANALYSIS AND NUMERICS, FACULTY OF MATHEMATICS AND COMPUTER SCIENCE, UNIVERSITY OF MÜNSTER, GERMANY

Email address: `ajentzen@cuhk.edu.cn`; `ajentzen@uni-muenster.de`